DEGREE PROJECT

Improving Internal Flow Logistics through Layout Design and Simulation-Based Validation

A Pre-Projection Study for a Future Logistics Centre at Alimak Group Sweden AB



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Abstract

This thesis investigated how Systematic Layout Planning (SLP) and Discrete Event Simulation (DES) could be applied to improve internal flow logistics at Alimak Group Sweden AB's production site in Skellefteå, Sweden. The study addressed inefficiencies in the current logistics configurations, including long internal transport distances caused by a poorly located goods reception, fragmented storages across multiple buildings, crossing material flows and outdoor packing activities that exposed workers to significant temperature variation in northern Sweden.

Systematic Layout Planning was applied as the primary design methodology, supported by participatory observations, interviews and brainstorming. The SLP process was expanded from five conceptual steps into ten operational steps to increase traceability and support integration with simulation-based validation. Seven layout alternatives were generated and weighted against the specification of requirements with a weighted decision matrix, covering flow efficiency, closeness relationships, space utilisation and safety and ergonomics. Layout 4 was selected as the preferred concept, achieving a total requirement fulfilment score of 92.2%.

The proposed layout consolidated all logistics functions within a single building, reducing key transport distances by 30-89% compared to the current state and replacing a multi-step inbound handling process with a single consolidated receiving flow supported by conveyor belts and Automated Guided Vehicles (AGVs). Discrete Event Simulation in Siemens Plant Simulation was used to validate the selected layout under operational conditions. All simulated product categories met or exceeded their weekly throughput targets, with industrial elevator packing reaching eleven units per week against a target of eight units, and no station experiencing material delivery delays during operational hours. No structural bottleneck was identified in the simulated model, though this finding should be interpreted in light of the model's simplified demand assumptions and scope.

The results confirmed the feasibility of the proposed logistics centre layout in meeting the target production rate of 20 units per week (industrial elevators and construction hoists combined) whilst improving safety, ergonomics and inventory transparency. Recommendations were provided for AGV fleet sizing, Automated Storage and Retrieval System (ASRS) integration, implementation sequencing and workforce transitioning to support the operational realisation of the proposed concept.

Keywords:
Systematic Layout Planning, Discrete Event Simulation, Material Flow Optimisation, Bottleneck Identification, Production System Design, Ergonomic Workplace Design

Sammanfattning

Detta examensarbete undersökte hur Systematisk Layoutplanering (SLP) och Diskret Händelsestyrd Simulering (DES) kan tillämpas för att förbättra den interna flödeslogistiken vid Alimak Group Sweden AB:s produktionsanläggning i Skellefteå, Sverige. Studien adresserade ineffektiviteter i den nuvarande logistikkonfigurationen, däribland långa interna transportavstånd orsakade av en felplacerad godsmottagning, fragmenterade lager fördelade över flera byggnader, korsande materialflöden samt packningsstationer placerade utomhus som exponerade arbetare för betydande temperaturvariationer i norra Sverige.

Systematisk Layoutplanering tillämpades som primär designmetod, med stöd från deltagande observationer, intervjuer och brainstorming. SLP-processen expanderades från fem konceptuella steg till tio operativa steg för att öka spårbarheten och stödja integrationen med simuleringsbaserad validering. Sju layoutalternativ genererades och utvärderades mot en kravspecifikation med hjälp av en viktad beslutsmatris, som täckte områdena flödeseffektivitet, närhetskrav, ytanvändning samt säkerhet och ergonomi. Layout 4 valdes som det primära konceptet med en total kravuppfyllnadsgrad på 92.2%.

Den föreslagna layouten konsoliderade alla logistikfunktioner inom en byggnad, reducerade viktiga transportavstånd med 30-89% jämfört med nuläget och ersatte en flerstegs inkommande hanteringsprocess med ett enda konsoliderat flöde för godsmottagning, skapat med transportband och Automated Guided Vehicles (AGV:er). Diskret Händelsestyrd Simulering i Siemens Plant Simulation användes för att validera den valda layouten under operativa förhållanden. Alla simulerade produktkategorier uppnådde eller överskred sina veckovisa genomflödesmål, där packning av industrihissar nådde elva enheter per vecka mot ett mål på åtta enheter, samt ingen station upplevde förseningar i materialleveranser under operativa timmar. Ingen flaskhals identifierades inom modellens djup, även om detta resultat bör tolkas med medvetenhet kring modellens förenklade antaganden gällande efterfrågan och avgränsningar.

Resultaten bekräftade genomförbarheten av den föreslagna layouten för att uppnå målet med en produktionstakt på 20 enheter per vecka (industrihissar och bygghissar kombinerat), samtidigt som säkerhet, ergonomi och lagertransparens förbättras. Rekommendationer ges för AGV-flottans storlek, ASRS-integration, implementeringssekvensering och personalövergång för att stödja det operativa förverkligandet av det föreslagna konceptet.

Nyckelord:
Systematisk Layoutplanering, Diskret Händelsestyrd Simulering, Materialflödesoptimering, Flaskhalsidentifiering, Design av Produktionssystem, Ergonomisk Arbetsplats

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1 Introduction

Inefficient material flows and long internal transports can significantly limit production performance in industrial environments. In response to these challenges, Alimak Group Sweden AB, referred to as Alimak, is planning a new logistics centre at its production site in Skellefteå, Sweden. This thesis explores how a systematic layout design, supported by Discrete Event Simulation can be used as an efficient and reliable logistics solution for future production.

This Master Thesis is conducted by M.Sc Student Johanna Holmlund as a mandatory final semester project on the M.Sc Industrial Design Engineering: Production Design program at Luleå University of Technology. The execution of this thesis is supported by supervisor Mattias Karlsson, Production Engineering Manager at Alimak as well as Therese Öhrling, Senior Lecturer at Luleå University of Technology.

1.1 Background

For over 70 years, Alimak Group Sweden AB, referred to as Alimak, has been a world-leading company in vertical transport for the construction and industrial sectors. Alimak manufactures both permanent and temporary elevators, construction hoists as well as working platforms. The company has a worldwide customer base and has an installed base of over 23 000 units (Alimak, n.d.). Alimak is located in Skellefteå and is also a part of the global Alimak Group. On site in Skellefteå, industrial and construction elevators are produced, and products can be delivered either as standard solutions or customized according to customer requirements. Today, Alimak has a production rate of 10 units per week with a utilisation rate of 80% at their stations. Alimak Group also has manufacturing and assembly facilities in several other countries, including China and other parts of Europe, where different components and vertical access solutions such as platforms are produced. The corporate headquarters of Alimak Group is located in Stockholm, Sweden.

Alimak has, like many other companies, suffered from nonoptimal layout planning in relation to expansion over the last 30 years. Because of this, the production site suffers from long internal transports, many small storage areas as well as unnecessarily long and crossing flows, resulting in inefficient product and material flows.

Alimak is planning to build a new logistics centre on their production site in Skellefteå, Sweden. The new building will contain and gather all handling of goods, excluding raw materials. The new building will also supply production with materials as well as work as the main storage for pallet storage, storage for larger parts on shelves and smaller ones in paternoster systems, where the main storage contains most of the material in manufacturing. All the material from the storage areas will be transported by Automated Guided Vehicles (AGVs) and forklifts. The new centre will also include storage for consumer goods as well as storage for cables. The logistics centre will be home for spare parts, including picking, packing and shipment. The new building will also include packing of elevators and elevator accessories as well as logistics for shipment. The logistics centre will also contain changing rooms for employees, a gym, a restaurant, conference rooms as well as a showroom. Today, a general layout for the logistics centre has

been created by the Production Engineering Manager at Alimak, which will lay the foundation for the development of the new layout, specifically the activities and connected areas included.

1.2 Assignment Description

Today, Alimak in Skellefteå is experiencing issues with internal transports, mainly because of the location of the goods reception, far from the manufacturing processes. The goods reception is also distant from the entrance on the premises, meaning the delivery trucks have to drive through the entire area to arrive at the goods reception. This is creating safety risks and an inefficient flow. Therefore, there is a need for a pre-projection to map the production flows and from this and the general layout, create an optimal layout for a new logistics centre that can meet the desired production rate. The new layout will then be strengthened and verified with Discrete Event Simulation (DES), to show the possible production rate as well as possible bottlenecks in the production. DES is used to evaluate the dynamic behaviour of the proposed layout under realistic conditions, enabling the ease of identifying bottlenecks and capacity limitations that are otherwise difficult to assess through physical layout analysis.

1.3 Purpose and Goal

The purpose of this thesis is to analyse the needs Alimaks future production has and based on this, create a decision basis for how the new logistics centre layout should be constructed to reach an efficient flow that will meet future demand. The thesis also aims to challenge current flows and processes to evaluate if the final, proposed layout can handle a production rate of 20 units per week.

The goal in this project is to execute a pre-projection that can be used as a basis when designing the new logistics centre. This shall result in an optimised layout that is visualized with a 3D simulation, where possible bottlenecks and limitations in the production flow can be identified.

1.4 Research Questions

Through the assignment description and choice of methodology, the following research questions have been formulated:

RQ1: What are the main sources of inefficiency and production losses in the current material flow connected to the new logistics centre at Alimak's production site in Skellefteå?
RQ2: How can the internal material flow and layout be redesigned to support an efficient logistics centre that supports a production rate of 20 units per week whilst maintaining a good work environment?
RQ3: How does the proposed logistics centre layout perform in terms of throughput, resource utilisation and bottlenecks when evaluated using Discrete Event Simulation?
RQ4: Which parameters have the greatest impact on the system performance in the simulated logistics centre?
RQ5: What are the strengths and limitations of combining SLP and DES as a methodology for pre-projection layout design in an industrial logistics context?

1.5 Stakeholders

Stakeholders in this project can be divided into internal and external stakeholders. The most affected internal stakeholders include the on site workers at Alimak in Skellefteå where a new logistics centre will impact their everyday operations and work life. Alimak Group will also be an internal stakeholder in this project since a new building will result in a significant financial investment. External stakeholders include customers, who will also be affected by the new logistics centre as well as collaborating partners, if the production rate increases. Suppliers will also be an external stakeholder and affected by a higher demand on material as well as delivery trucks who will need to be informed by the new location of arriving goods, separated from the delivery of raw materials.

1.6 Limitations

Limitations in the project included the time frame available for the thesis project, which consists of 30 hp (högskolepoäng/ECTS credits), equivalent to 800 hours over a 20 week period, from January 19th 2026 to June 7th 2026. In this project, a pre-projection refers to a conceptual and analytical study to support future decision making, rather than a detailed investment plan or implementation study. The project is also limited within the production site at Alimak. Processes located in other buildings on the premises, such as manufacturing, assembly and supporting functions are excluded from the scope of this study. The layout design is limited to the activities performed within the new logistics centre. Areas of the building not associated with logistic activities, such as the restaurant, changing rooms and the showroom, are excluded from the scope of both the layout design and the simulated model.

2 Theoretical Framework

This chapter presents a theoretical foundation for the thesis, which is a framework structured around interconnected domains. Production system design and layout methodology, material flow and internal logistics, simulation based evaluation and industrial safety and human factors. Safety and ergonomics theory ensures that the socio-technical dimension of the design is addressed. Warehouse efficiency literature provides additional context for the operational performance goals of the new facility.

2.1 Production System Design and Production Development

Production systems consist of interconnected technical, organisational and human elements that together enable value creation. The design of a production system therefore involves more than arranging equipment or defining process steps. It encompasses decisions regarding process structure, material flow, resource allocation, control principles and spatial configuration. Within this broader system perspective, layout design constitutes a central structural decision, as it defines how material, information and people interact physically within the system (Bellgran & Säfsten, 2010).

Layout decisions affect not only transport distances and efficiency but also ergonomics, worker safety exposure and operator interaction conditions, reinforcing the need to treat spatial design as a human factors problem, rather than a purely technical one (Abad, 2018). This is particularly relevant for the logistics centre at Alimak, where both AGVs and forklifts operate in close proximity to workers performing picking, packing and assembly-adjacent tasks.

According to Bellgran and Säfsten (2010), production development can be described as a structured process for designing or redesigning production systems in order to meet new strategic, technological or operational requirements. When an organisation moves into a new facility, layout development becomes a core production development activity. The relocation of existing stations into a new building requires reconfiguration of spatial relationships while preserving functional dependencies between processes. Furthermore, Bellgran and Säfsten (2010) emphasise the importance of long-term flexibility and adaptability. A new facility layout should therefore not only support current operational requirements but also allow for future changes in volume, product mix or technology. This perspective is directly applicable to the logistics centre project at Alimak, where the target output of 20 units per week represents a deliberate increase beyond current operational capacity and where the spatial configuration must accommodate potential further growth.

2.2 Material Flow and Internal Logistics Design

Material flow efficiency is widely recognised as a critical performance factor in production and logistics systems. Efficient material flow reduces transport effort, waiting times and work-in-progress (WIP) levels, thereby improving overall system performance (Bellgran & Säfsten, 2010). Poor material flow often leads to bottlenecks, increased handling, unnecessary backtracking and resource idling.

Jiang et al. (2023) emphasise that logistics and material flow problems differ from classic facility layout problems because the cost of movement often dominates operational cost. When transport lines are long

or complex, material handling cost can represent a large share of total system cost. This insight is directly relevant to the current situation at Alimak, where goods reception is located at the far end of the premises, requiring delivery trucks to cross the entire site, and thereafter material go through multiple internal transport steps before reaching their point of use. The layout of the new logistics centre is therefore designed explicitly to align spatial arrangement with dominant material flows, reducing transport work and improving responsiveness.

In manufacturing and logistics research, poorly structured material flow patterns are associated with increased congestion, cross-traffic and bottleneck formation. Since bottlenecks often emerge from interactions between processes rather than isolated capacity limitations, flow design must be evaluated at the system level (Negahban & Smith, 2014). In Alimak's case, the fragmentation of storage and packing functions across multiple buildings creates crossing flows and repeated handling steps, which both increase transport per order and introduce safety exposure. A consolidated logistics centre with coherent flow logic directly addresses these structural inefficiencies.

Lean principles, which originated in manufacturing but have been widely adopted in logistics and internal material handling contexts, emphasise waste elimination and flow improvement (González-Reséndiz et al., 2018). In internal logistics, waste typically takes the form of unnecessary transport, waiting or excess material handling. Layout design should therefore aim to minimise unnecessary backtracking and intersecting transport routes (Bellgran & Safsten, 2010). From this perspective, the new logistics centre is designed to eliminate cross-building transfers, consolidate storage within a single structure and position picking, packing and shipment activities in close spatial proximity.

According to Jiang et al. (2023), facility layout design determines how physical spaces, stations, storage areas and transportation paths are arranged within a production or logistics system. Efficient layout design directly affects material handling costs, movement distances, utilisation in the system and overall operational performance. In systems where logistics intensity is high, layout planning must consider both facility placement and transportation pathways to minimize unnecessary handling and travel distances. Tetik (2025) further notes that a large share of project time and cost in logistics-intensive environments is related to material handling and deliveries, reinforcing the strategic importance of spatial decisions for operational efficiency.

2.3 Systematic Layout Planning

Systematic Layout Planning (SLP) is a structured approach for developing and comparing layout alternatives based on relationships among functions, activities and material flows. SLP guides layout designers through steps such as requirement analysis, relationship evaluation and layout concept creation, making the process transparent and repeatable for both block layouts as well as detailed layout planning (Muther & Hales, 2015). According to Bellgran & Säfsten (2010), structured layout methodologies are particularly valuable during conceptual production system design because they enable traceability between functional requirements and spatial solutions.

However, SLP produces a static layout representation and does not inherently evaluate time-dependent performance metrics such as waiting times, resource utilisation or transport congestion. As Negahban &

Smith (2014) highlight, static approaches may be insufficient in environments characterised by stochastic variability, congestion and resource interaction. On the other hand, simulations are particularly suited for capturing dynamic interactions, variability and resource interaction. Therefore, while SLP structures the conceptual design process in this project, it is complemented by Discrete Event Simulation to evaluate dynamic system behaviour. The combination of these two methods reflects the recommendation in the literature that structural design and dynamic evaluation reinforce each other in production system development (Bellgran & Säfsten, 2010).

2.4 Industrial Safety and Human Factors in Layout Design

Industrial safety is a critical consideration in internal logistics systems, particularly where transport equipment operates in proximity to personnel. Production system design decisions influence work conditions and risk exposure (Abad, 2018), reinforcing the socio-technical perspective of integrated human-technical systems. Bellgran and Säfsten (2010) argue that safety should be treated as an inherent design parameter in production system development rather than as an operational afterthought. Accordingly, safety considerations are incorporated into layout relationship analysis and spatial allocation.

Industrial safety in logistics systems concerns the prevention of accidents and risks arising from interactions among people, vehicles and materials. Layout design plays a crucial role in safety outcomes by defining transport routes, interaction zones and space maneuvering. Internal transportation zones, where forklifts or AGVs operate among pedestrians represent high-risk areas for incidents if not properly planned (EU-OSHA, 2001). At Alimak, in the current state, delivery trucks must drive through the premises, creating interactions with internal transport equipment and pedestrian movement. This represents a concrete and avoidable safety risk that the new logistics centre is specifically designed to eliminate.

Ergonomic considerations are closely related to safety in internal logistics contexts. Ergonomics-based studies demonstrate how safety and performance are intertwined in design-decisions, reducing unnecessary transport, minimising manual handling of heavy goods and ensuring appropriate workspace dimensions all contribute simultaneously to worker wellbeing and operational efficiency (Abad, 2018). In the current state at Alimak, elevator packing is performed outdoors in northern Sweden, exposing workers to significant temperature variation across seasons. Relocating these activities indoors within the new logistics centre addresses safety exposure and work environment.

2.5 Warehouse Efficiency

Warehousing is a central component of logistics systems and has a substantial impact on both operational efficiency and total logistics cost. Prior research shows that warehouse performance is strongly influenced by facility layout and design, inventory management, material flow, handling operations as well as shipping and distribution processes (Chen, 2025). Inefficient layouts, poor inventory control and unnecessary material handling increase non-value-adding time, energy consumption and labor requirements, thereby reducing overall system performance.

From a facility planning perspective, warehouse design decisions are therefore shown to influence throughput, space utilisation and operational flexibility. Chen (2025) demonstrates that optimising warehouse layout, including storage system selection, aisle configuration and positioning of receiving and shipping areas, has measurable effects on both cost and capacity. In the context of the logistics centre at Alimak, the current storage configuration is fragmented across multiple buildings and systems, limiting inventory visibility and generating replenishment loops that increase internal transport frequency. The new facility consolidates these functions within a single building and employs a structured storage strategy, including an ASRS system for medium sized goods, paternoster systems for small parts and conventional shelving for large parts - to improve both throughput and stock transparency. The storage system design also has direct implications for AGV integration. When storage locations are logically grouped and spatially consistent, AGV routing becomes more predictable and efficient. Fragmented storage increases the complexity of automated transport coordination and raises the risk of congestion at storage interfaces. The warehouse efficiency literature therefore supports the SLP-driven emphasis on storage clustering and spatial coherence as a prerequisite for effective automated internal logistics.

2.6 Discrete Event Simulation as a Concept Verification Tool

Discrete Event Simulation (DES) is an analytical tool used to model and study systems where changes in state occur at discrete time points when events happen, such as arrival of goods, resource releases or completion of tasks. DES captures the interactions among entities, resources, queues and processes to replicate real-world process behaviours under varying conditions (González-Reséndiz et al., 2018).

DES is particularly suitable for systems characterised by variability in arrival rates, processing times and resource interactions. Simulation allows evaluation of performance indicators such as throughput, resource utilisation, queue lengths and bottleneck locations (Negahban & Smith, 2014). Negahban and Smith also emphasise that simulation is widely used in manufacturing system design to evaluate alternative configurations prior to physical implementation. By capturing dynamic interactions, DES reveals performance effects that static distance-based models cannot identify. This makes DES a natural complement for the static representation created using SLP.

Simulation-based approaches provide a means to address limitations of static layout analysis by enabling evaluation of logistics capacity alignment prior to construction. By modelling material inflows, on-site handling and outbound flows in relation to the production schedule and site layout, simulations allow different scenarios to be tested and bottlenecks to be identified in advance. In this sense simulation functions as a verification tool, assessing whether planned logistics capacities are sufficient to support the intended production plan and site layout under realistic conditions (Tetik, 2025).

In this project, DES is applied after the SLP process to validate performance of the final layout concept. The purpose is verification rather than comparative optimisation where the selected layout is tested under realistic operating conditions to determine whether it meets the required throughput of 20 units per week without excessive congestion or over-utilisation of resources. Key performance indicators include throughput, resource utilisation, queue lengths, lead times and potential bottleneck locations. The use of simulation as a risk-reduction tool in this context is consistent with recommendations in production design literature (Bellgran & Säfsten, 2010).

2.7 Summary and Theoretical Synthesis

The areas of theory presented in this chapter together form a coherent analytical foundation for the thesis. Material flow and internal logistics theory defines what the layout is optimised for, which is reduced transport intensity, eliminated backtracking and minimised handling steps. This provides the conceptual tools to identify inefficiency in the current state.
Industrial and human factors theory ensures that the socio-technical dimension of the design is not sacrificed for throughput gains. Therefore separation of vehicle and pedestrian flows, indoor relocation of packing activities and ergonomically considered workstations and placements are treated as design requirements alongside flow efficiency. Warehouse efficiency research provides operational context, linking storage system selection and layout coherence to measurable outcomes in throughput, space utilisation and inventory visibility.

Taken together, these theoretical domains establish that the quality of a logistics centre layout cannot be assessed by any single criteria. Flow efficiency, safety, ergonomics, storage logic and dynamic system behaviour must all be considered simultaneously. This integrated perspective shapes both the methodology applied in Chapter 3 and the evaluation criteria used to assess layout alternatives in Chapter 5.

3 Methodology

This chapter explains the different methodologies that will be used during the project. This includes the methodology for the structure of the project, the methods for planning and designing the layout as well as methods that will be used for verification of the layout by using DES.

3.1 The Project Spiral

In this thesis project, the method used throughout the whole process is The Project Spiral (Ranhagen, 1995). The method, which is shown in *Figure 1*, consists of a circular, six step process repeated three times to create a spiral, where revisits are made to each phase every lap and the results are reworked. The phases of The Project Spiral are the following:

1) Plan for change.
2) Run diagnosis.
3) Formulate goals and demands.
4) Search and evaluate.
5) Choose an option.
6) Develop and work through details.

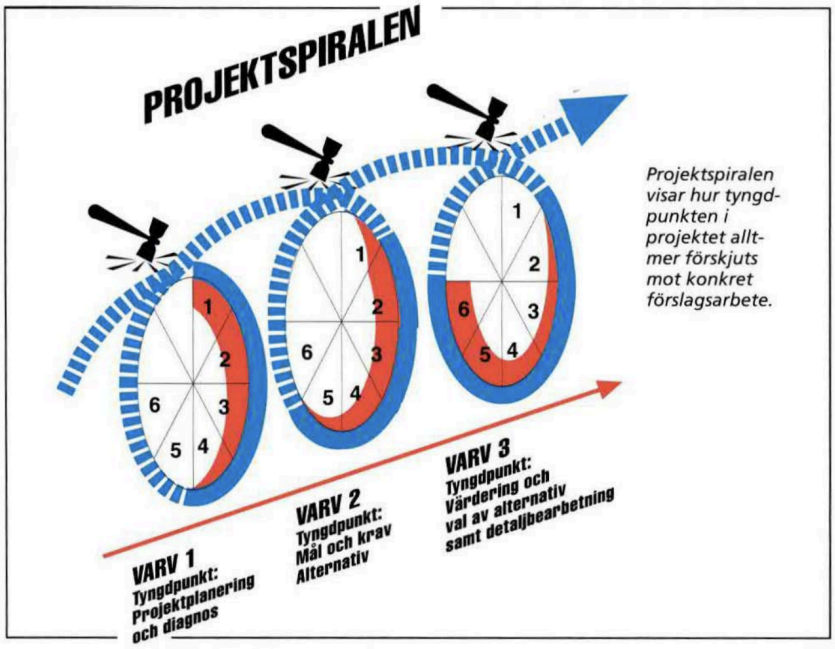


Figure 1: The Project Spiral (Ranhagen, 1995).

The three laps all focus on different phases of the circle, where the first lap focuses on planning and running diagnosis, the second lap focuses on goals, demands and different alternatives. The third and final lap has a clear focus in weighing the different alternatives against each other to select the most performative as well as detail work. The breakdown of the Project Spiral divided onto the weeks of the project can be seen in *Appendix 1*.

3.2 Systematic Layout Planning

Systematic Layout Planning (SLP) is conducted by following a framework of phases that the layout project passes through. This includes procedures for planning, to identify, visualise and to rate concepts, relationships and activities. The phases include:

I) Location: *Where the area is to be laid out.*
II) General Overall Layout: *Deciding on the general arrangement of the laid out area.*
III) Detailed Layout Plans: *Placing each part of equipment and machinery.*
IV) Installation: *Plan for installation and get plan approved.*

In this project, the location has already been determined, which means that this project will start from phase two. This project is also a pre-projection which means that installation is outside of this project's scope. Main focus will therefore be on phase two followed by phase three. In *Figure 2*, a capsule summary of SLP can be seen.

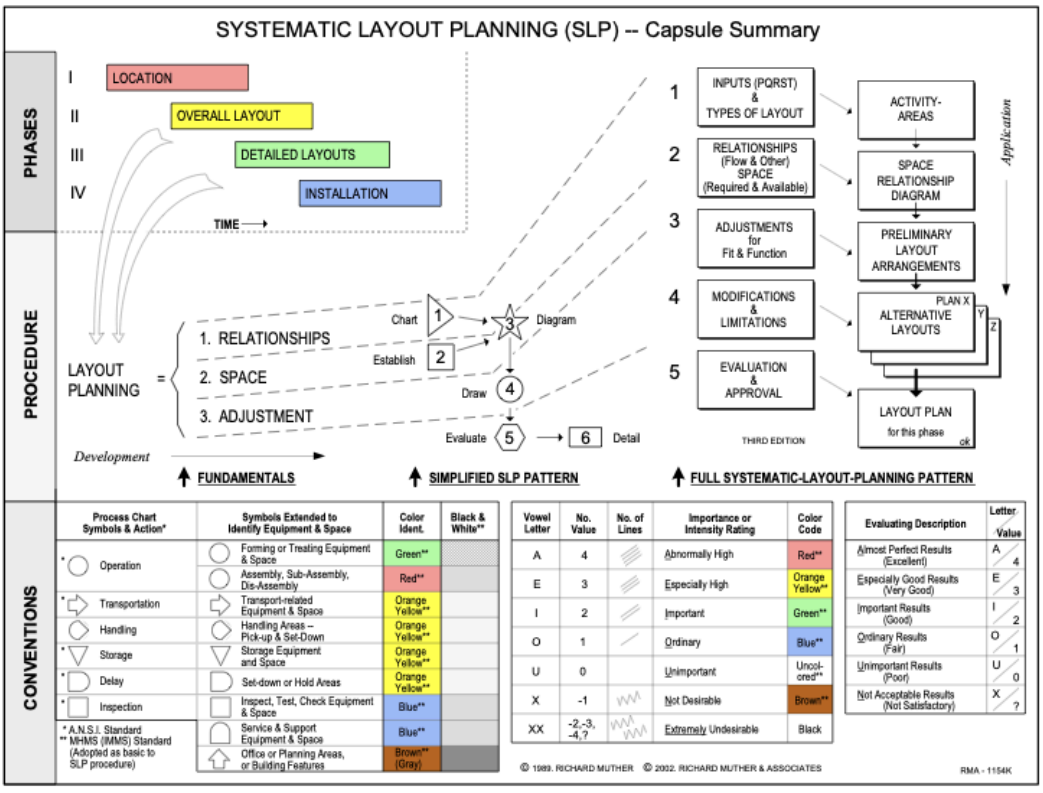


Figure 2: Capsule Summary of Systematic Layout Planning (Muther, R. & Hales, L., 2015)

According to Muther & Hales (2015), Phase II and Phase III are carried out through a set of five conceptual steps, which are the following:

1) Inputs (PQRST) & Types of Layout
2) Relationships & Space
3) Adjustments for Fit & Function
4) Modifications & Limitations
5) Evaluation & Approval

To increase clarity and applicability, these five conceptual steps have been modified and expanded into ten operational steps for this project. This structure allows each activity to be treated explicitly and supports the validation process through Discrete Event Simulation later in the process. The modified SLP structure will be presented in the following section.

3.2.1 Phase II: General Overall Layout

Step 1: System definition (PQRST)

The first step of the Systematic Layout Planning (SLP) consisted of defining the system using the PQRST framework (Muther & Hales, 2015), Through the framework, the Products, Quantities, Routing, Supporting Services and Time could be found for the logistics centre at Alimak, establishing the operational foundation of the layout problem.

Products referred to the materials and components handled within the logistics centre, including incoming goods, internally handled goods and finished assemblies. Quantities described the volume, frequency and flow intensity of these goods. Routing defined the sequence of processes and internal movements, while supporting services included utilities, handling equipment and related logistical activities necessary for operation. Time captured production rhythms, delivery frequency and operational scheduling.

This step also included creating a table that described the objective, scope and constraints for the logistics centre. The objective describes what the purpose of the SLP is, while included processes show the activities inside the logistics centre. System boundaries are limitations connected to the SLP, which mainly states that the SLP will only be applied to the logistics centre and not to other production buildings on site. This boundary was necessary to prevent suboptimization, where local improvement negatively influenced upstream or downstream operations. Additionally, project constraints were systematically documented. Building constraints describe the shape and size of the building and how safety constraints are connected to general industrial safety principals. Lastly equipment and process constraints were determined.

Step 2: Material and Product Flow Analysis

The second step of the SLP was to analyse material and product flow within the logistics centre. For this part, an Operation Process Chart (Muther & Hales, 2015) was made. The chart showed how materials flow inside the logistics centre. The analysis determined the most effective sequence of material flow with the necessary processes involved. The chart also showed where activities are connected with other parts of the production, such as final assembly, located in a different building. The purpose of this analysis was not only to document process order but also to identify high frequency interactions, critical transport links and external dependencies (e.g. final assembly in another building).

Step 3 and 4: Identification of Activities and Space Requirement Analysis

The third and fourth step were combined with the purpose of simplifying presentation of findings in one single table. This step included defining each activity within the building that requires space, as well as estimating the space needed. To estimate space required, comparisons were done with the current layout and measurements for each activity and estimations regarding if more space was needed for a higher production rate. The supporting layout created by Mattias Karlsson was also used as support to estimate space requirements. This step ensured that spatial demand was systematically identified before positioning activities relative to each other.

Step 5: Activity Relationship Analysis

The fifth step was to evaluate the closeness needed between all activities by using a relationship matrix with the following ratings:

- A: Absolutely necessary
- E: Especially important
- I: Important
- O: Ordinary closeness
- U: Unimportant
- X: Undesirable

The purpose of this step was to determine relationships and which activities need to be prioritized in closeness within the logistics centre. The Relationship Chart (Muther & Hales, 2015) not only shows activities that have a relation to each other but also gives a rating of importance and reason for why the relation is necessary. By documenting the reasoning behind proximity requirements, the basis for later spatial decisions was made transparent.

Step 6: Development of Space Relationship Diagram

Step 6 included creating a Space Relationship Diagram that shows both the relationship importance as well as the space requirements for each activity. This is done by creating blocks of the correct size for each activity and by starting with the most important relations, connecting the activities with lines with a specific length and colour that represents the desired closeness. This diagram provided a visual representation of how activities could be positioned relative to each other while simultaneously respecting both spatial constraints and prioritised relationships.

Step 7: Generating Layout Alternatives
Based on the Space Relationship Diagram, different layout alternatives were created by using various flow patterns, such as straight-through, U-flow, L-flow, Comb-flow and combinations thereof. Alternative input and output points were also tested within existing constraints to examine how material entry and exit locations influence internal flow patterns. Different aisles orientations were explored, as well as options for combining or separating activities. Generating multiple alternatives allowed comparison of different structural solutions rather than relying on a single configuration.

Step 8: Evaluation and Selection of Layout Concept
The layout alternatives were then evaluated by using a predefined specification of requirements. Each alternative was assessed based on how well it fulfilled the defined requirements. By systematically comparing the alternatives against the same criteria, the selection process was able to maintain structure. Following this evaluation, one layout concept was chosen for further development.

3.2.2 Phase III: Detailed Layout Planning

Step 9: Detailed Layout Planning
For Step 9, detailed layout planning was done to determine precise placements of workstations, equipment, storage areas, buffers and transport areas. This also included defining material handling areas, safety zones and designated paths for vehicles and pedestrians.

For larger functional areas, such as the spare parts area, packing of hoist accessories and packing of elevators, the detailed layout was carried out by repeating step 1-8 in the SLP, with necessary modifications. This ensured internal consistency within larger blocks. For smaller activities and areas, the layout decisions were made in consultation with employees at Alimak, allowing practical operational knowledge to inform final positioning.

Step 10: Layout Validation Through Discrete Event Simulation
The selected layout was then simulated using Discrete Event Simulation in order to analyse output, bottlenecks and resource utilisation. The simulation provided a dynamic evaluation of the layout under operational conditions, allowing potential performance limitations to be identified prior to possible implementation.

3.3 Participatory Observations

During the project, one week was dedicated to participatory observations in selected production areas that will be included in the future logistics centre. The purpose was to get an in-depth understanding of current material flows, work processes and routines. Wikberg Nilsson, Ericson, & Törlind (2021) describes the method as a way to identify undiscovered areas of improvement and to identify needs for the user in specific situations. Participatory observations are described as a way to understand the user's situations by watching, listening and asking questions. Wikberg Nilsson, Ericson & Törlind also describe that the observations should be done to gain insight and understanding of important aspects to consider that are difficult to gain only through interviews. It also gives context to a certain task, by how it is connected to others as well as attitudes the user has in the actual situation.

The observations were conducted through active participation of daily work tasks, allowing a first hand understanding of tasks, material handling and interactions between different activities. Particular attention was given to material flow paths, transport distances, space requirements, handling methods, equipment, waiting times, interruptions and non-value-adding time. In addition, informal discussions with workers during the observation period provided insights into employee experiences, perceived bottlenecks and suggestions for improvement. The observations were documented through notes and were used in activity identification for the SLP as well as the for the Operation Process Chart.

3.4 Interviews

Interviews were conducted with four selected employees representing different departments relating to the logistics centre, which was one employee at spare parts, two at the goods reception and one at elevator packing. The purpose of the semi-structured interviews was to complement the observations made and obtain a broader understanding of challenges, requirements and expectations connected to the new logistics centre. Wikberg Nilsson, Ericson & Törlind (2021) describe interviews as a way to meet the user and to understand their experience, opinions, attitudes, motivations and behaviour connected to your work. Interviews can be used in different stages and a varying amount of times depending on the purpose of the interviews. In the beginning of a process it can be used as a way to gain understanding of their knowledge, typical behaviours in the workplace, what parts of certain processes they find useful and what irritates them. Later in the project, interviews can be used as a way to involve the user in the design process and to gain their thoughts and opinions about different concepts.

A guide for the interviews ensured consistency through all interviews, whilst still allowing interviewees to express their experiences and perspectives. The interviews focused on material flow challenges, interactions with different departments, space and layout related issues, perceived inefficiencies as well as overall suggestions for improvement in the future logistics centre. Interview results were summarised and used as support when executing the current state description as well as performing the SLP. The interview questions can be seen below and the results of the interviews can be seen in the *Appendix 2, 3 and 4.*

1. Can you briefly describe your role and your main responsibilities?
2. How long have you worked in the production at Alimak?
3. Can you describe how material moves in this department from when it arrives until it is sent onward?
4. Where do you experience that material often has to wait or is moved unnecessarily?
5. Are there any flows that you find particularly complicated or inefficient?
6. Are there any workstations or functions that you think should be located closer to each other?
7. Are there any functions that absolutely should not be located near each other? Why?
8. Where do you experience that material shortages often occur?
9. Where does too much material or congestion occur instead?
10. Are there any buffers that work well today and should be kept?
11. Which tools and equipment do you use most in your work?
12. Do you feel that the transport equipment is sufficient today?
13. Are there situations where transport causes disruptions or waiting?
14. Are there places where you experience safety risks related to transport or layout?
15. Are there any situations where pedestrians and transport vehicles cross each other in an unsuitable way?
16. If you could change one thing about the layout, what would it be?
17. What do you think is the most important for the new logistics centre to solve?
18. Is there anything in today's way of working that absolutely must be kept?
19. Is there anything else you think is important to consider that we have not talked about?

3.5 Brainstorming

As a part of the layout work, both the block layout and for the detailed layout planning, brainstorming was the main methodology used to generate a wide range of alternatives before narrowing down the most feasible solutions. The purpose of using brainstorming at this stage was to encourage divergent thinking and avoid prematurely locking the design into a single concept. Since layout decisions have long-term operational consequences in terms of material flow efficiency, safety, flexibility and scalability, it was important to first explore multiple structural configurations before evaluating them. Structured brainstorming sessions were conducted where different layout alternatives were created using sketches, scaled drawings and digital tools. Wikberg Nilsson, Ericson & Törlind (2021) describes the method as a procedure to generate a large amount of ideas by focusing on big and crazy ideas, without criticism. Combinations of different ideas are also common, where quantity over quality is the focus.

During the block layout phase, brainstorming was used to explore macro-level configurations. This included testing different flow principles (e.g. U-flow, circular flow and comb flow) and testing various placements of key functional areas such as goods reception, storage zones, packing areas and shipping. The goal in this stage was not precision but conceptual clarity and comparison between different layout logics. In the detailed layout planning phase, brainstorming was more focused and constraint-driven. The alternatives were refined by incorporating limitations such as safety regulations, AGV aisles within activities, storage system specifications, work area placements and AGV dropping zones.

3.6 Specification of requirements

To evaluate how well different layout alternatives met the requirements of the new logistics centre, a specification of requirements was created. The specification defines key requirements and criteria reflecting both operational needs and strategic objectives. The development of the requirement specification was based on findings from the current state analysis, stakeholder input, production forecasts and input from supervisor Mattias Karlsson. The purpose was to translate qualitative project goals into measurable evaluation criteria that could systematically guide decision-making during concept selection. The template for the specification of requirements can be seen below, in *Figure 3*.

Code	Demand	Weight
F1	Minimise transport distances between sequential processes	4
F2	Support high transport frequency without bottlenecks	4
F3	Reduce risk of congestion in transport routes	4
F4	Minimize material handling effort and number of handling steps	5
C1	Activities with A/E/I relationships are positioned in close proximity	4
C2	Undesirable adjacencies (X relationships) are avoided	3
C3	The layout supports logical process sequencing	5
C4	Connections to external departments are maintained efficiently	4
SU1	Layout fits within building geometry and structural constraints	4
SU2	Provides adequate space for all activities and supporting services	5
SU3	Allows accessibility to workstations, storage and transport areas	4
SU4	Enables flexibility for future production rate increases or layout adjustments	3
SE1	Separation of pedestrian and vehicle flows where possible	5
SE2	Minimised conflict points between transport routes and work areas	4
SE3	Provides sufficient visibility in transport corridors and intersections	5
SE4	Ensures access for emergency routes and safety zones	5
SE5	The layout considers work environment and minimizes physical load	5
Total (%)		

Figure 3: Specification of Requirements Template

Each requirement was assigned a weight based on its relative importance to the overall project objectives. The weighting process was carried out to distinguish between critical needs and desirable attributes, ensuring that essential operational functions were prioritised in the evaluation. Requirements directly affecting operational continuity, safety and capacity were generally given a higher weight compared to secondary optimisation criteria.

3.7 Weighted Decision Matrix

The evaluation of layout alternatives was performed using a weighted decision matrix. According to Wikberg Nilsson, Ericson & Törlind (2021), a weighted decision matrix is a way to evaluate and rank concepts based on predefined criteria, in this case the specification of requirements. The criteria are descriptions of traits in the concepts that should be met. The purpose of the method is to rationalise the concepts with a starting point as a way to define to what extent a criteria is met, rather than personal opinions of the different ideas (Wikberg Nilsson, Ericson & Törlind, 2021). Each concept was assessed against every requirement and assigned a score reflecting the degree to which the requirement was

fulfilled, between 1 and 3. The score was then multiplied by the corresponding weight, resulting in a weighted value. By summing the weighted values for all requirements, a total percentage score was calculated for each alternative.

This structured evaluation enabled transparent and objective comparison between concepts. Rather than relying solely on qualitative judgement, the weighted assessment provided a traceable decision-making framework. The alternative with the highest overall degree of requirement fulfilment was selected for further development and detailed analysis. However, the quantitative result was also complemented by engineering judgement to ensure that practical feasibility and system robustness was adequately considered.

3.8 Discrete Event Simulation

Discrete Event Simulation (DES) has been used as a method for validation of the chosen concept in order to verify production rates, resource utilisation and potential bottlenecks within the system. For this project, Siemens Plant Simulation was used to build a three-dimensional model to evaluate the system's capacity.

DES is applied after the SLP process to validate the performance of the final layout concept. The purpose of using DES in this project is validation rather than comparative optimisation. Rather than modelling multiple competing layout alternatives, the selected concept is tested under realistic operating conditions to determine whether it meets the required performance targets. Some changes can be made to the model to accommodate the findings.

3.8.1 Scope and Level

The simulation model covers the internal logistics operations of the proposed logistics centre, beginning from the point where materials are already in storage and ending when finished goods are transferred to the external staging area awaiting shipment. Shipping itself is not included in the model boundary.

The model is built on an operational level of detail, capturing material handling events, resource allocation and queue formation, without modelling the physical geometry of the building as well as each individual elevator component. All materials are represented using material units (MUs) that represent products from each storage area. This reflects a real operational constraint at Alimak, where detailed information about incoming goods, such as how quantities are distributed across pallets or how much physical space a given shipment occupies, is not available in a structured form. Introducing distinct MUs for every individual component type within each storage system would therefore require a large number of unfounded assumptions and would not increase model realism.

Activities included within the simulation boundary are retrieval and transport of materials from the four storage systems, including the ASRS for standard pallet-based goods, the paternoster system for small parts, the shelving for large parts and the dedicated cable storage to their respective workstations. From there, the model covers spare parts picking and packing, packing of hoist accessories and packing of elevator cages, doors and panels as well as packing of extra material, through to outbound staging for shipment.

Activities explicitly excluded from the model boundary include inbound goods reception, external truck arrival rates and docking logistics, manufacturing processes in adjacent buildings, consumer goods storage and distribution, usage and distribution of packing material, final assembly of elevator cages and personnel facilities. The construction hoists from the final assembly do not require any packing and are transported directly from final assembly to external storage and are therefore not included in the packing flow within the model. Machine breakdowns and unplanned stoppages were not modelled, consistent with the validation objective of assessing steady-state performance under normal operating conditions.

3.8.2 Input data, distributions and assumptions

Input data for the simulation was derived from three sources, including historical data extracted from Alimak's IFS Business System, through discussions with workers and supervisors as well as through assumption-based estimates for parameters where empirical data was unavailable or insufficient.

The target production rate of 20 units per week was used as the basis for outbound flow frequency, converted to a daily rate assuming five operational days per week under standard single-shift scheduling, with 12 of these units being construction hoists and eight industrial elevators. Since construction hoists do not pass through the packing flow, the effective demand driving the elevator packing station is eight units per week.

Processing times at workstations were modelled using triangular distributions. These were selected because of their suitability for representing bounded, asymmetric process time data in situations where the sample size is insufficient to fit more precise parametric distributions with statistical confidence (Law, 2015). Triangular distributions require only three parameters, a minimum, a most likely and a maximum processing time, which were estimated through production supervisor Isabelle Lindgren, supported by data from the IFS business system.

AGV travel times were estimated to a speed of 1.5 m/s and loading and unloading at each AGV drop zone were assumed to follow a constant handling duration of 2 minutes per interaction.

3.8.3 Validation and Verification Approach

Since the model represents a system that does not yet exist, traditional validation against a current-state operational baseline is not possible. Instead, validity was ensured through an incremental build process in which each subsystem was constructed and tested independently before being integrated into the full model. Logical behaviour was confirmed at each stage by verifying that entities followed expected routing, that queues formed and cleared as anticipated and that resource interactions produced plausible behaviour. The overall model was monitored through the build process to detect unexpected behaviour.

3.8.4 Experimental Setup

A warm-up period of one simulated day was applied before statistics collection began. The warm-up period was determined by visual inspection of work in progress (WIP) levels across the model, confirming that the system had reached steady state behaviour, characterised by material being present at

all stations and AGV activity being continuous, before the measurements window opened. This approach follows the method recommended by Law (2015) for simulation with steady state objectives.

The observation period for each replication was set to 31 days of simulated time, yielding 30 days of collected data and covering 23 working days under the assumed shift schedule. This duration was selected to capture sufficient order cycles and AGV replenishment events to produce stable mean estimates for throughput and utilisation KPIs. A run length of at least ten times the expected system cycle time is considered adequate for steady state estimation (Negahban & Smith, 2014), and this criteria is met by the chosen run length.

Five replications were performed for each experimental scenario. All reported statistics represent means averaged across five replications, accompanied by 95% confidence intervals. Key performance indicators collected include throughput per product category, station utilisation and identification of potential bottlenecks through queue length monitoring.

4 Description of current state

This chapter presents a structured analysis of the current logistics configuration at Alimak's production site in Skellefteå. The purpose is not only to document the existing layout and material flows, but to identify structural characteristics that affect transport effort and distance, handling frequency, safety exposure and operational transparency. The chapter is organised into organisational context, product characteristics, layout structure and flow analysis, followed by a synthesis of identified inefficiencies.

4.1 The organisation

Alimak Group Sweden has approximately 330 employees on site. The production site in Skellefteå primarily manufactures industrial and construction elevators where both standardised elevators and customer-specific configurations are produced. The majority of production activities are performed in-house, while selected operations, such as manufacturing of selected electrical components, surface treatments such as galvanisation and welding of stainless steel are outsourced.

Due to the high level of product customisation, production flows vary between orders. Different elevator types require different components, dimensions and processing steps. As a result, material flow and routing are not fully standardised, which increases the complexity of internal logistics planning. This variability implies that logistics must support fluctuating routing patterns and material combinations, increasing dependency on storage flexibility and transport responsiveness.

The main production activities on site include processing of raw materials (e.g. fiber laser cutting, bending and welding), surface treatment and paintwork, assembly of elevator cages, assembly of doors and flooring, final assembly, electrical components, packing of elevators and accessories and spare parts handling and distribution. Logistics support all of these processes through material reception, storage, internal transports, picking and outbound handling.

The logistics function connects goods reception, storage areas and production processes. All incoming material is first received centrally before being distributed to the appropriate storage or production area. Internal logistics therefore plays a critical role in ensuring material availability and production continuity. Since production is spread across multiple buildings, logistics also functions as a link between physically separated stations. This creates a strong dependency between layout structure and transport efficiency. Consequently, spatial configuration directly influences the amount of internal transport required to sustain production flow.

4.2 Products and materials

Alimak receives both raw material for internal processing as well as purchased components, such as bolts, screws and springs. In addition, consumer goods, returned goods, outsourced components and packing materials are delivered to the site. All incoming materials, regardless of final usage location, are first received at goods reception. From there, materials are transported either to storage areas or directly to production stations.

Since Alimak produces customer-specific elevators, component variation is high. Different elevator sizes and configurations require different parts and dimensions. This increases storage diversity and handling complexity, as a broad range of materials must be available simultaneously.

The material structure can be categorised into high frequency small components, pallet-based and half pallet-based materials and large parts exceeding pallet dimensions. This diversity increases the need for multiple storage systems and contributes to fragmented storage locations across the site.

Packaging conditions are also highly varying, both for goods reception and shipment. For goods reception, goods can be in small cardboard boxes, with orders of a small amount of springs for example, to medium deliveries on pallets with multiple hundreds of certain products. It can also be large components that may exceed pallet dimensions, where tractors or other special handling equipment are required. For deliveries, the shipping of elevators can also vary to a great extent from the smallest version with the measurement 395x156x165 cm, but can reach measurements as big as 435x303x235 cm. Shipping of spare parts on the other hand, are smaller but can still vary. Certain spare parts are small and can be manually handled, whilst others require a forklift or tractor to transport.

The variation in packaging size and handling requirements results in fluctuating space demand, variable transport equipment usage and differences in handling time.

4.3 Current layout and production flows

The current layout at Alimak is divided into multiple buildings on the premises. *Figure 4* shows the overall layout and relevant buildings for the project. The relevant parts of the production have been marked in the figure and parts where production that is not relevant to the project has been left blank. The buildings on the top of the figure show half of the reception and office building on the right hand side and the changing rooms, showers and restaurant on the left. The green lines in the figure illustrate transport areas.

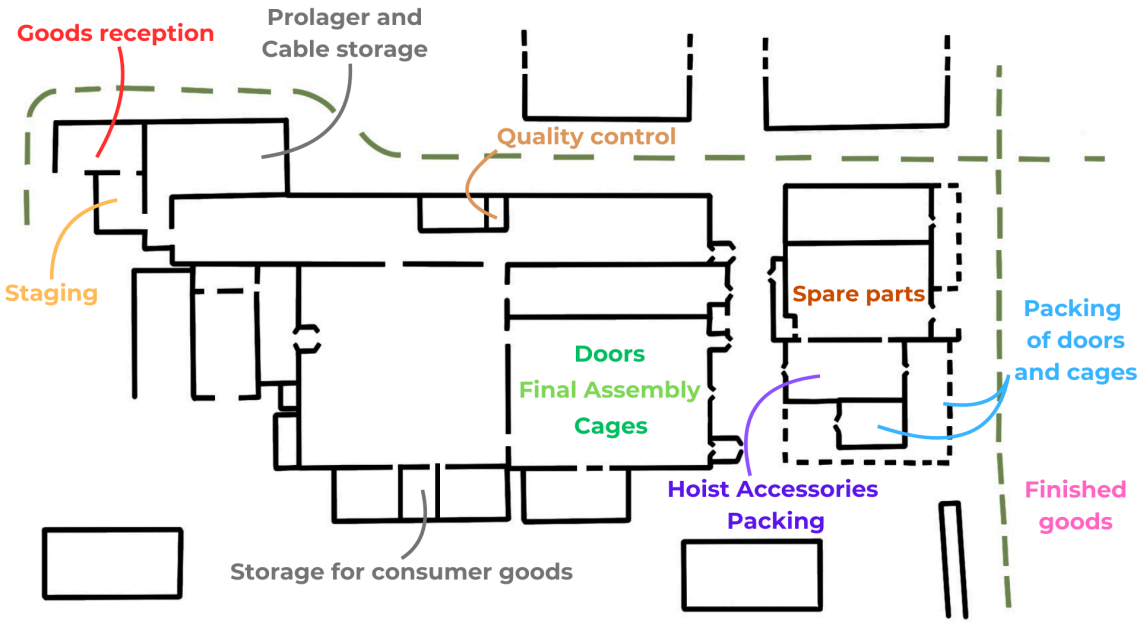


Figure 4: Overall Layout of Alimak Today

The goods reception is located at the far end of the premise, see *Figure 4*. Delivery trucks must drive through the site to reach this location. Adjacent to the goods reception is the main storage, referred to as "Prolager". This storage space includes storage of standard pallets, half pallets, storage space for bulky material as well as paternoster storage systems for smaller parts. The paternoster systems are connected to the Kanban system that Alimak uses throughout their production. Next to Prolager, is the cable storage and cable handling area, which includes storage shelves for finished cable orders and raw cable materials. *Figure 5* shows in greater detail the current layout of the goods reception, staging, Prolager and the cable storage and handling.

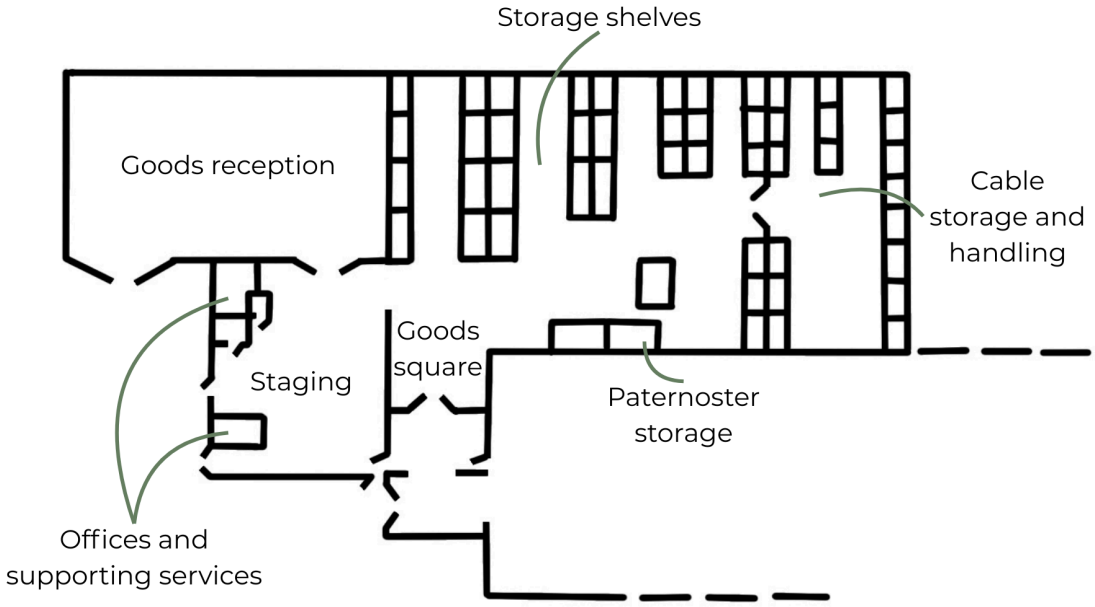


Figure 5: Detailed Layout of Goods Reception and Storage

Figure 6 shows the Process Flow Chart of the goods reception and describes the current flow, where goods are received and unloaded outside. The goods are thereafter placed in the goods reception or to an external storage area, meant for bigger parts that can not be stored inside. The goods that are placed in the goods reception area are thereafter transported to the staging area by forklift, where the material is added in the digital business system, IFS. A paper is then printed out that assigns the material to a storage space. The workers also open the boxes and check that the contents and the amount is correct. The goods then get sorted into pallets depending on which storage it will be stored at. The goods will then be transported to the goods square and thereafter moved into the storage system. If a quality control is necessary, it will be done before the goods are moved to the goods square.

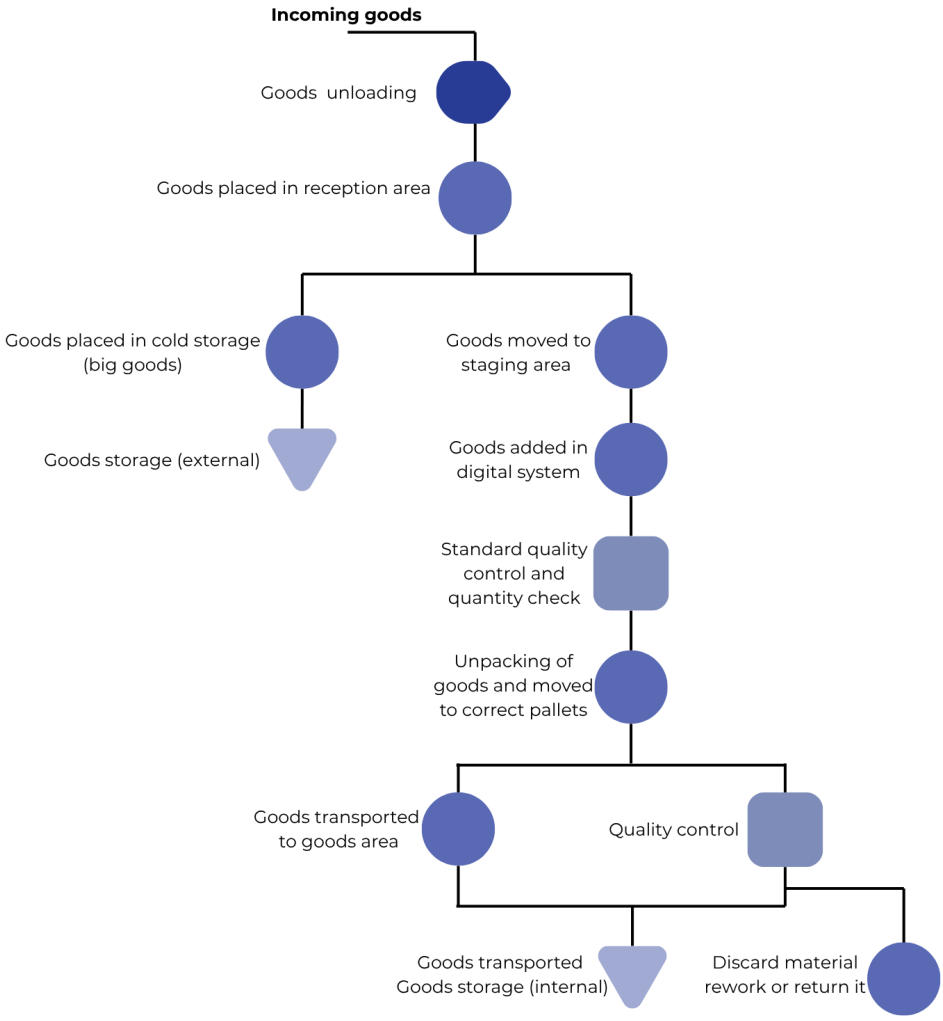


Figure 6: Process Flow Chart for Goods Reception

The quality control department is centrally located in the production area, see previous *Figure 4*. Quality inspectors are assigned to different areas as needed, to look at different materials and quality of processes, in areas such as welding and goods reception. A Coordinate Measuring Machine (CMM) is used by the inspectors.

For spare parts, elevator assembly and packing of hoist accessories, electrical components are needed. These are manufactured in the same building as the quality control. Today, there is a common issue with a limited availability of electrical components, which can create production delays. Within this building, the storage for consumer goods is also located. The material is delivered in the goods reception and then transported to the storage for consumer goods. Thereafter it is distributed as needed within the production.

The final assembly of elevators is done in the central building, see *Figure 4*. From the final assembly, the elevators are transported outside to the building adjacent to the final assembly (the building that will become the logistics centre). The industrial elevator cages are then packaged outdoors in wooden crates. The construction elevator cages do not require any extra packaging and are transported directly to the area for finished goods. Doors for elevators are transported from the final assembly of doors, and then packaged in the same area as the cages as well as in the adjacent building along with the required panels. In this area, extra material can also be packed onto pallets when needed. This can be extra material the customer requires or parts needed that do not fit into any other crates or boxes. Any additional material needed for packing of elevators are transported to the area, or picked by workers at elevator packaging. The packing of hoist accessories is done inside, by the same workers that do the packaging of elevator cages and doors. One person is responsible for the packaging of one entire elevator and chooses in which order to do it, depending on the availability at a certain station. The hoist accessories are grabbed by workers from different storage spaces, mainly their own storage next to the packing and Prolager. When a picking order is placed, the workers at Prolager will get an order and then transport the necessary materials to the packing of hoist accessories. When the packing is done, it is transported to the outside storage area for finished goods, placed next to the finished elevator cage.

The spare parts picking and packing is located in the same building as the packing of hoist accessories, see *Figure 4*. The logic is the same as for packing of hoist accessories, where a picking order is placed and materials needed from different storage areas, such as Prolager is transported to the facility. At the spare parts picking and packing, they also have their own storage system, where the Kanban system is used. Refills are made from Prolager. When orders are picked and packed, they are transported to the shipping area, where shipping labels are printed. An order for shipment is placed and the parts are stored until delivery. A detailed layout of the current building can be seen in *Figure 7*. Packing of hoist accessories include material that is needed for the assembly of the elevator, when the customer receives it. This can include cables for the mast section, bolts and screws, drive units and similar.

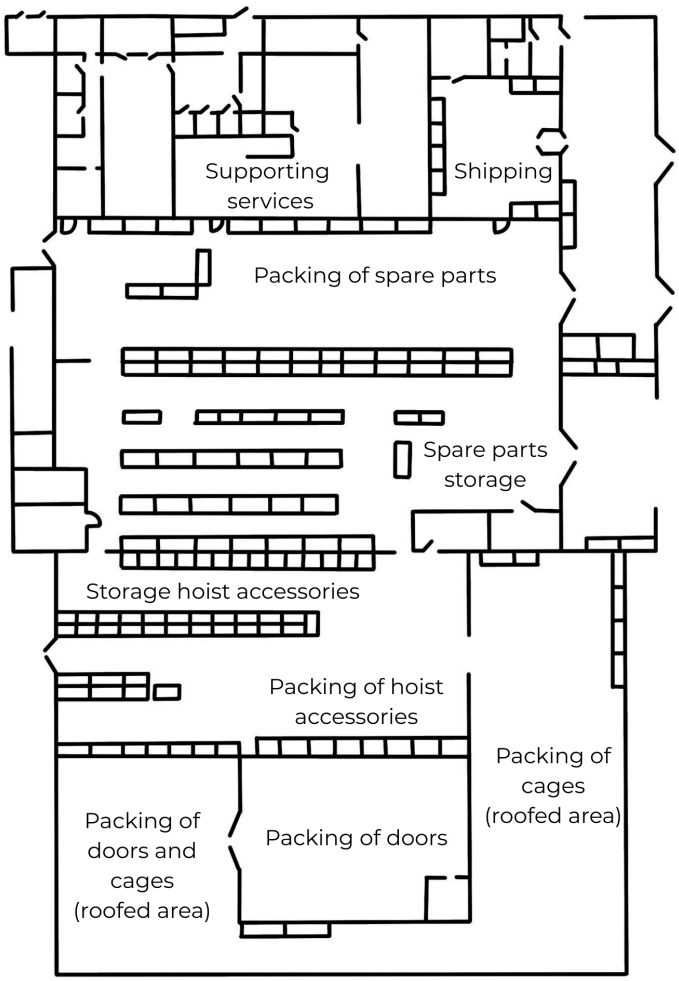


Figure 7: Detailed Layout of Current "Logistics Centre"

Figure 8 shows a Process Flow Chart for the current spare parts picking and packing. When an order is placed, a signal is sent to the main storage facility, where the required parts are gathered and transported to the spare parts department. Simultaneously, the relevant goods are picked from the spare parts departments' own internal storage. Once all parts have arrived, an appropriately sized box is selected.

Box selection cannot be done in advance, as the system does not store dimensional data for the parts. This means the worker must physically inspect all items before choosing the correct box size. The parts are then packed and taken to a weighing station, after which the box is moved to the shipping area and placed on shelves to await delivery booking. Once a delivery is booked, the box is retrieved from the shelf, a shipping label will be put on and it is returned to the shelf until delivery.

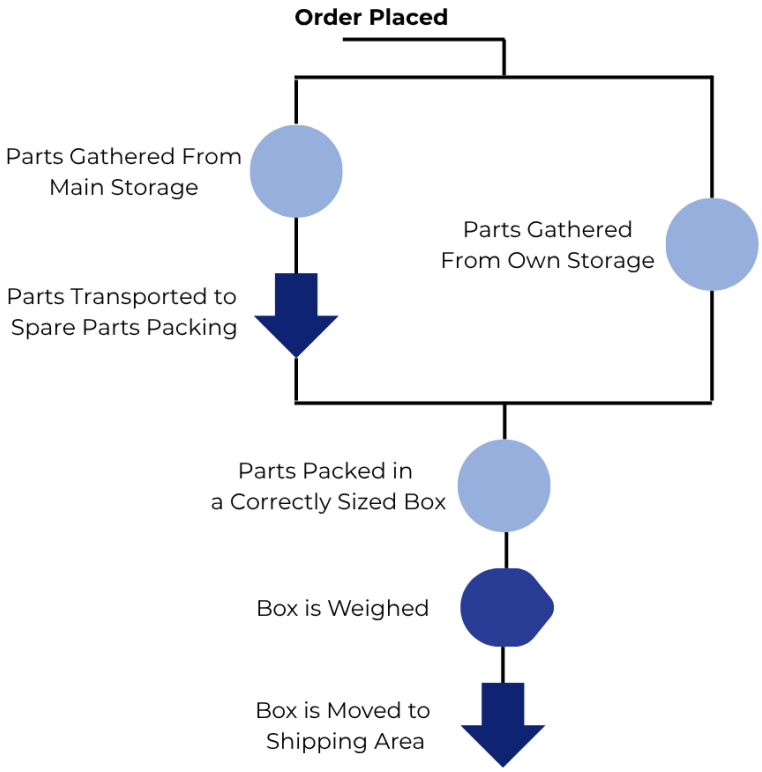


Figure 8: Process Flow Chart for Spare Parts Picking and Packing

Figure 9 shows the Process Flow Chart for packing of elevators, hoist accessories, doors, panels and extra material. All parts of a packing order are done by the same person, which is why the packing processes are shown in one figure. When an order is placed, material from the local storage at elevator packing is gathered, material from outside storages, both in other buildings and from outdoors, as well as a crate to pack in. All of these materials are packed as hoist accessories for an elevator order. Thereafter, the doors and panels are packed in a crate, followed by extra material packed on pallets. Lastly, the elevator is packed in another crate. All the different crates and pallets are then stored outside in the finished goods storage. Since one worker is responsible for an entire order, in what sequence the different packing steps are done is up to the individual worker, and can depend on availability at certain stations and in materials.

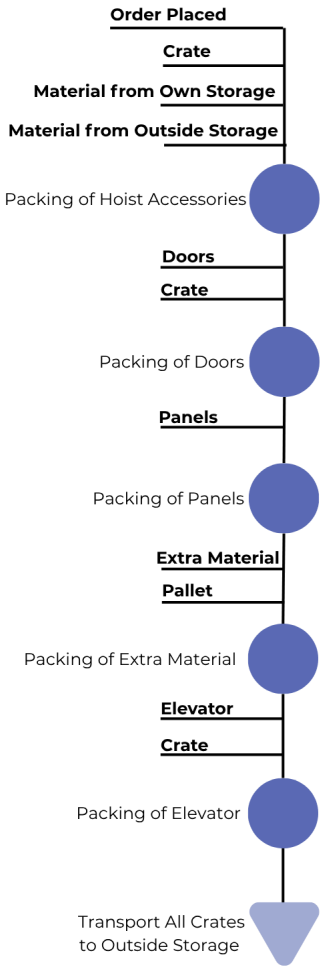


Figure 9: Process Flow Chart for Hoist Accessories, Doors and Cages

Overall, storage, packing and assembly functions are distributed across buildings, requiring logistics to compensate for physical separation through transport activities.

The above mentioned parts of the production at Alimak are the processes that are included in the new logistics centre. Beyond this, the manufacturing of the elevator cages, from raw material to finished goods is also done, as well as doors and flooring. Within this, welding, laser cutting, bending, paintwork, assembly, safety mechanisms and such are also made. Since these processes are not included in the scope of the project, no detailed description of the processes will be provided.

4.3.1 Distances in the current site layout

An overview of the current site layout can be seen in *Figure 10*. The overview illustrates the goods reception location, main storage (Prolager), production buildings, final assembly and spare parts picking, packing and shipment. The layout highlights the physical separation between buildings and the distances between key logistics functions. As *Figure 10* depicts, goods reception is located far from the final assembly and packaging areas, approximately 220 metres. As a result, internal transport is required across long distances. Delivery trucks must drive 350 metres through the premises, creating interactions with internal transport and pedestrian movement.

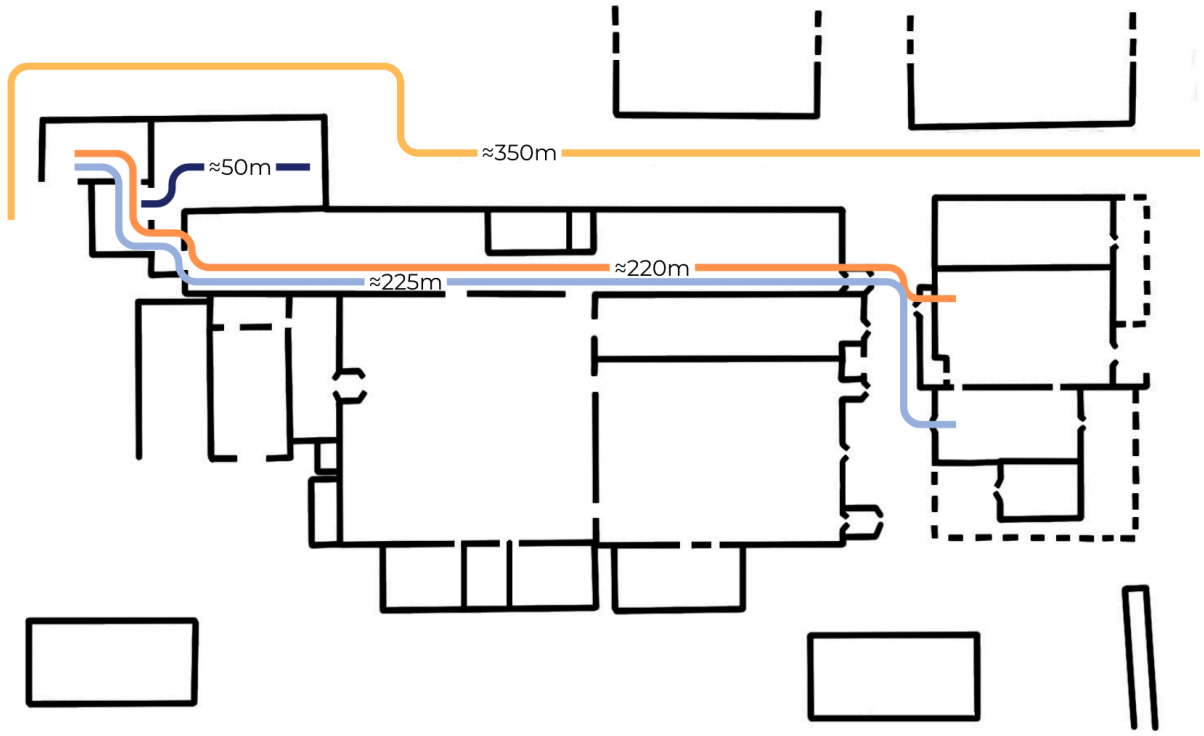


Figure 10: Most Determining Distances in Current Layout

4.3.2 Current inbound material flow

In *Figure 11*, the current inbound material flow is depicted. The figure shows the transport made by the delivery truck, which is represented by the blue line. Thereafter, the green line shows the transport made by workers at the goods reception, normally using a forklift, to the staging area and thereafter to the goods square. Three of the following lines thereafter show the transports into Prolager, where the purple line illustrates the transport to storage shelves, the grey line shows the transport to the cable storage and the light blue line shows the transport to the paternoster storage systems. The red line shows the transport made to quality control and the orange line to storage for consumer goods.

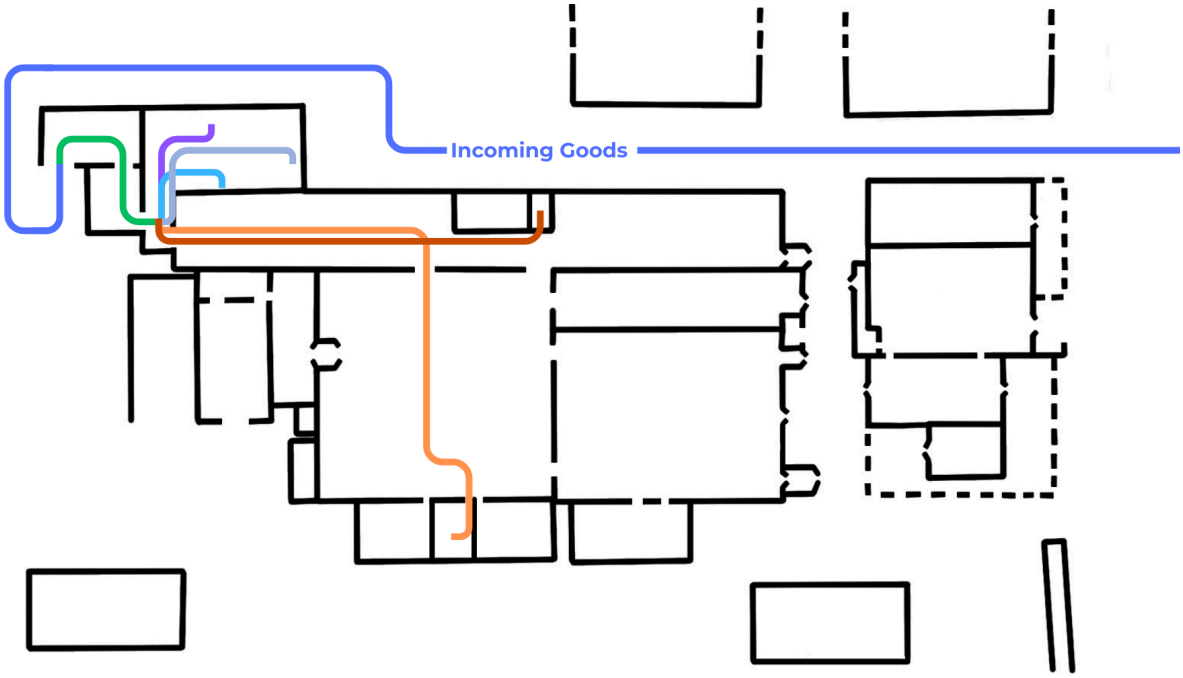


Figure 11: Current Relevant Inbound Material Flows

Each additional handling increases lead time, labour requirements, risk of damage, safety risk and ergonomic risks. Internal transport routes cross pedestrian paths and production areas, increasing the exposure to safety risks even further. Depending on destination and storage location, a single inbound order may require between two and five handling operations before reaching its point of use. This indicates a relatively high transport intensity per material unit.

4.3.3 Internal material flows relevant to the future logistics centre

Since the future logistics centre does not include production processes or manufacturing, the material flows connected to the logistics centre is limited to flows that consider handling and storage. The logistics centre will therefore include goods reception, main storage, spare parts picking and packing, packing of hoist accessories, cage and doors packing and storage for consumer goods.

In *Figure 12*, the current distribution of material relevant to the logistics centre can be seen. This includes transports from the main storage to spare parts picking and packing as well as packing of hoist accessories.

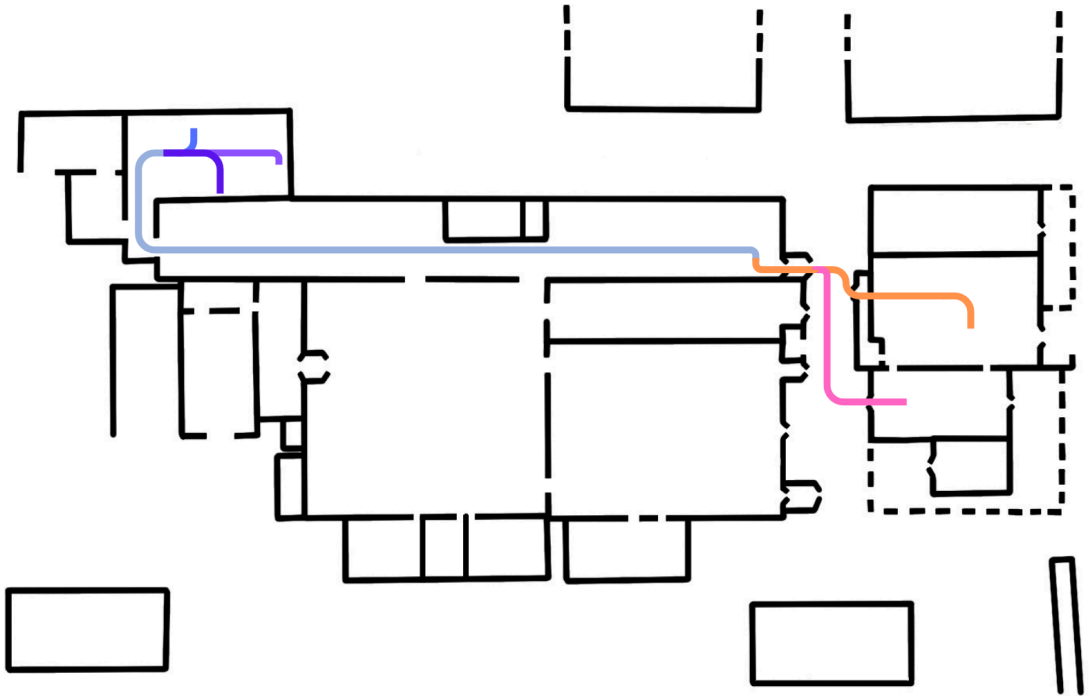


Figure 12: Current Relevant Distribution from Prolager

Figure 13 depicts the outbound material flow, that is elevator cages, doors and accessories that are finished as well as spare parts ready for shipment. The orange line shows the spare parts flow. The purple line shows the flow for the elevator cages, the pink line shows the hoist accessories and lastly the green line shows the elevator doors and panels.

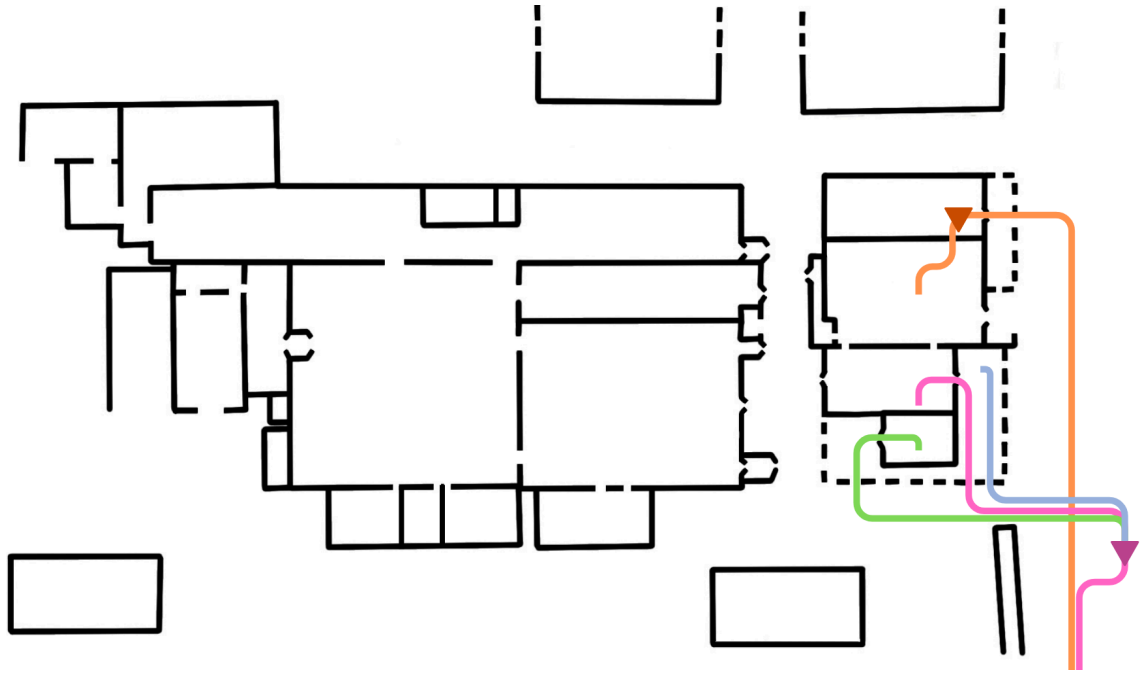


Figure 13: Current Relevant Outbound Material Flows

4.3.4 Identified problems and inefficiencies

A significant portion of material handling involves transportation between buildings and different storage spaces. Since all materials are received at goods reception and subsequently distributed to various storage areas or directly to production, at least one internal transport is required for all inbound material. In many cases, additional transports occur when materials are later moved from storage to production or packaging areas. This structure increases transport per order, number of handling operations and labor tied to non-value-adding activities. Repeated handling of materials also increases the risk of damage and delays, while also contributing to higher operational workload.

The current layout creates multiple crossing transport routes. Materials are transported across the site in different directions depending on destination and in several cases, pass through or around other operational areas. In addition, certain materials are moved back and forth between buildings due to the separation of storage and usage locations. This indicates that spatial configuration does not fully align with process flow, resulting in unnecessary movements. Crossing flows also increase the risk of congestion, collisions between forklifts and pedestrians and a complexity in the internal traffic management. These conditions therefore introduce safety risks and reduce flow efficiency.

Storage areas are divided across several buildings and systems. Certain materials are stored in multiple locations due to proximity requirements, while others require replenishment from Prolager. This fragmentation leads to additional replenishment transports, reduced transparency of inventory distribution and increased coordination complexity. When storage is not centralized or logically grouped, material availability becomes harder to monitor and balance. This can contribute to delays or temporary shortages, particularly in areas dependent on specific components, such as electrical parts.

The presence of multiple storage systems and physical locations reduced overall visibility of stock levels. The structure complicated inventory tracking, stock balance management and identification of shortages. Limited overview increases the risk of material not being available when needed, which may result in production delays or inefficient prioritisation decisions.

Packaging of elevators is currently performed mainly outdoors. This exposes workers to weather conditions, where in northern Sweden temperatures between -30 degrees Celsius in the winter and +30 degrees Celsius in the summer are not uncommon. It also introduces additional internal transport between final assembly and between the different packing areas for elevators and its accessories.

Overall, the current layout reflects incremental development rather than coordinated system design. Functions that are logically connected in the production process are physically separated, requiring compensating transport activities. The logistics function therefore spends considerable effort compensating for spatial separation rather than supporting streamlined flow. This misalignment between process logic and spatial configuration results in increased internal transports, repeated handling, higher safety exposure and reduced operational transparency.

5 Future state mapping

Based on the inefficiencies identified in Chapter 4, this chapter presents the structured development of a new logistics centre layout. The objective is to reduce transport intensity, minimise repeated handling, improve inventory transparency and enhance safety by aligning spatial configuration with dominant material flows. The development follows the principles of Systematic Layout Planning (SLP) and is organised according to Phase II (General Overall Layout) and Phase III (Detailed Layout Planning).

5.1 Systematic Layout Planning

Systematic Layout Planning (Muther & Hales, 2015) was used as the primary methodology for developing the future logistics centre layout. The selection of SLP was motivated by the need to ensure traceability between identified inefficiencies and spatial design decisions, thereby reducing reliance on intuitive placements and instead grounding the layout in documented flow relationships and operational requirements.

5.1.1 Mapping of Phase II: General Overall Layout

Step 1: System definition (PQRST)

The first step in the project, which was a part of phase two, regarding the execution of SLP, was to find the PQRST for the logistics centre, which, according to Muther & Hales (2015), means identifying the products, quantities, routing, supporting services and time. Products refers to the goods produced in the area, which can be raw materials, purchased parts, treated parts or finished products. Quantity describes how much goods are produced, used and supplied. Routing is the sequence of the process with its supplies and operations and supporting services refers to the utilities or related activities that are essential for the area. Lastly, time involves when the products will be produced. The description for each of these factors PQRST, in relation to the new logistics centre can be seen in *Table A*. In the project, these were found by discussions with the Production Engineering Manager Mattias Karlsson at Alimak.

Table A: PQRST for the Logistics Centre

Factor	**Description**
P - Products	Spare parts, hoist accessories and packed elevator cages doors from assembly.
Q - Quantity	Target throughput of 20 packed hoists per week with spare parts handled continuously depending on demand
R - Routing	Incoming Goods → Unpacking → Quality Control → Storage → Assembly → Packing → Delivery
S - Supporting Services	Forklifts, ASRS System, tractors, AGV, receiving, shipping, toilets, office, garbage room, storage areas, pedestrian walking area, forklift and AGV driving area.
T - Time	Production adapted to the current production calendar, 06:30 - 15:30 with break from 08:00-08:15, lunch from 11:00-11:30 and a break from 14:00-14:15

In contrast to the current state, where functions are spatially dispersed, the system definition for the new logistics centre explicitly assumes functional integration within a single building. The assumption directly addresses the cross-building transport inefficiencies identified in Chapter 4.

The target throughput of 20 packed hoists per week represents an increase compared to current effective capacity. Therefore, spatial allocation must not only replicate current functionality but support higher material handling frequency without proportionally increasing internal transport effort.

Thereafter, a table that describes the objective, scope and constraints for the logistics centre was created, see *Table B*. The objective describes what the purpose of the SLP is, while included processes show the activities inside the logistics centre. Intermediate storage and transports are not included in the processes, but are a part of the project as well. System boundaries are limitations connected to the SLP, which mainly states that the SLP will only be applied to the logistics centre and not to other production buildings on site. Building constraints describe the shape and size of the building and safety constraints are connected to general industrial safety principals. Lastly equipment and process constraints are determined nonexistent.

Table B: Objectives, Scope and Constraints for the Logistics Centre

Category	**Description**
Objective	Design a new logistics centre that allows efficient production flow and supports a production rate of 20 units/week.
Included processes	Incoming goods handling, unpacking, quality control of incoming goods (if required), spare parts picking and packing, shipment of spare parts, storage areas, packing of tested hoists, packing of hoist accessories, lifting-out area with tractors, cable warehouse and storage until customer delivery.
System boundary	The study is limited to processes physically located within the new logistics centre. Processes performed in other buildings of the production site are out of the scope of this thesis.
Building constraints	New building with rectangular shape, dimensions 68m x 53.5m (approximately 3 638 m^2). Station "Packing of elevators" is constrained in the building as well as some industrial doors and regular doors.
Safety constraints	No predefined safety constraints at this stage, general industrial safety principals are assumed.
Equipment and process constraints	The location of elevator packing is constrained, to be compliant with the location of the final assembly of elevator cages in the adjacent building.

The clear definition of system boundaries prevents sub-optimisation by ensuring that layout decisions inside the logistics centre are aligned with upstream and downstream interfaces (final assembly and shipment). This is particularly important given that spatial misalignment was identified as a primary inefficiency in the current state.

Step 2: Material and Product Flow Analysis

Next in the SLP was to analyse material and product flow within the logistics centre. For this part, an Operation Process Chart (OPC) (Muther & Hales, 2015) was made. The chart showed how material would flow inside the logistics centre, see *Figure 14*. Compared to the current inbound flow, the proposed routing consolidates goods reception, quality control, storage and picking activities within one coherent spatial structure. This reduces the number of cross-site transport steps and eliminates cross-building transfers. The chart also shows where activities are connected with other parts of the production, such as final assembly, located in a different building. Material from the storages will also be needed in other buildings on the premises.

The OPC shows a simplified structure of the goods reception, where incoming goods will be received and thereafter unpacked. The goods will then go through a quality control (either a check from workers or a more intricate inspection) and will then be transported to the correct storage system. Goods transported to the consumer goods storage will be distributed in all buildings, which is shown through a simplification in the OPC. In reality, this would be connected to the whole production, but in *Figure 14*, it is described in text instead. When an order is placed, the relevant storages will pack kits with the needed material and thereafter an AGV will pick up the kits and transport them to the relevant packing station. These stations include packing of hoist accessories, packing of extra material and packing of spare parts. For extra material, parts can also be needed from an outside storage. For elevator packing, the elevator is transported from final assembly and the crate to pack in is brought inside, which are stored outside. Packing of doors and panels follow the same logic. Doors are delivered from the final assembly of doors and panels are stored next to the packing station for easy access. Crates are also stored outside and brought in when needed. After packing the finished combined kits, specifically hoist accessories and extra material, packed elevator as well as doors and panels are transported to the outside storage for finished goods. The packed spare parts are transported to the storage for finished goods, located internally. The storages can also send kits and goods into the production, when needed, shown in the OPC.

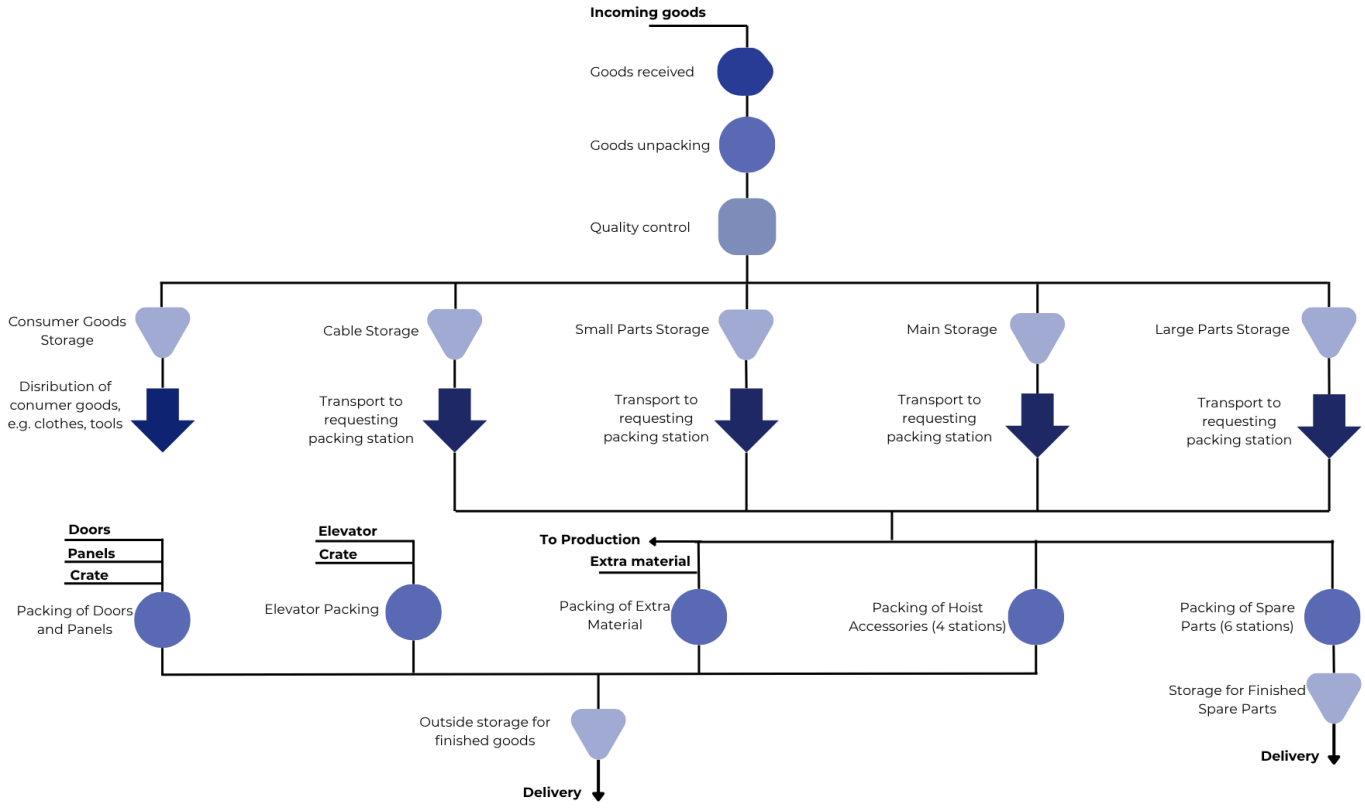


Figure 14: Operation Process Chart for the Logistics Centre

Step 3 and 4: Identification of Activities and Functional Areas and Space Requirement Analysis
In this step, all activities that are logistics related functions that directly impact the material flow and have space requirements were listed, see *Table C.* With each of these activities, an estimation of required space was also made. Personnel facilities and supporting services are seen as fixed constraints and are therefore not included in this activity analysis, but will be included in *Step 9* of the SLP.

Step four was to make an approximation of space required for each activity. Since the objective of the project is to increase production rate, space requirements are not based on proportional scaling of the building, but on the expected increase in material flow handling frequency and intermediate storage needs. Functions directly affected by higher throughput such as storage and staging areas are therefore allocated additional space, while other functions are dimensioned based on operational requirements rather than available floor area. AGV, forklift and pedestrian aisles are not included.

The inbound area was set to 686 m^2, reflecting the current size. This is justified by the transition to AGV-based material handling, which eliminates multi-step transports and significantly increases handling efficiency. The existing goods reception has demonstrated the capacity to support a production rate of 20 units per week under more standardized product conditions, which supports this claim.
Storage space allocations were determined in collaboration with Mattias Karlsson, based on current drawings of the storage areas. The design intent is to minimize inventory levels rather than expand storage capacity. Picking and packing reflects a slight area increase compared to current space allowances, with the deliberate exception of the elevator packing area which is enlarged to accommodate both the increased production rate and the requirement to conduct all packing activities indoors. Outbound areas also mirror the current state, structured around a stricter delivery precision standard and a stronger JIT philosophy. AGV and forklift charging stations are sized based on the confirmed number of vehicles required to operate the logistics centre. External areas are intentionally held to current allowances, consistent with the commitment to lean inventory management.

Table C: Identification of Activities with Space Requirements for the Logistics Centre

Activities	**Basis**	**Space requirements**
Inbound		
Goods reception	Receiving dock, staging, offices	511 m^2
Unpacking	Two/three unpacking stations	125 m^2
Incoming quality control	One inspection station, inspection office, shelves for pallets	50 m^2
Total Inbound		**686 m^2**
Storage		
Main storage	Kardex MLog System	380 m^2
Storage - small parts	Four paternoster storage units	8,3m^2 each = 33m^2
Storage - large parts	Shelves for large pallets	135 m^2
Storage for consumer goods	One paternoster storage system, shelves, cabinets	150 m^2
Storage for shipment	Shelves for pallets	70 m^2
Returned items	Shelves for returned items	30 m^2
Total Storage		**798 m^2**
Picking & Packing	X working stations and packing areas	
Picking & Packing		263 m^2
Packing of elevator accessories	X working stations and packing area	260 m^2
Packing of elevators	Working area for one elevator	70 m^2
Cable handling	Cable storage and cable handling	80 m^2
Total Pick and Pack		**673 m^2**
Outbound		
Parts shipment	Area for staging and shipping	90 m^2
Total Outbound		**90m^2**
Support Logistics		
Charging Stations for forklifts and AGVs	Area for chargers	60 m^2
Waste handling	Waste bins for different trash	27 m^2
Total Support Logistics		**87 m^2**
External Areas (fixed)		
Received goods (large)		300 m^2
Hoist accessories (large)		300 m^2
Finished elevator cages	Cages waiting for shipment	375 m^2
Finished hoist accessories	Accessories waiting for shipment	300 m^2
Elevator shipment	Loading onto truck	200 m^2
Total External Areas		**1475 m^2**

Step 5: Activity Relationship Analysis

For the next part, a relationship diagram was created based on the activities with space requirements, see *Figure 15.* The figure shows how each activity that was listed in *Table C*, should be placed in relation to each other. The closeness is determined by the A-E-I-O-U-X system, defining which level of closeness that is necessary, from Absolutely necessary to not desirable. For each of these closeness needs, a reason has also been determined and is shown by the numbers next to the letters in each box in the diagram. 1 relates to closeness needs because of material flow whilst 2 is for general convenience, 3 for movement of equipment and lastly 4 which connects to work environment.

	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1
Goods reception 1	U	U	I1	O2	O3	U	O2	X4	X4	X4	X4	X4	E2	U	U	U	I1	I12	A12	
Unpacking 2	U	U	U	E2	U	U	O2	U	U	U	U	U	U	I2	I1	I1	E1	E1		
Incoming quality control 3	U	U	U	I2	U	U	U	U	U	U	U	U	O1	U	U	U	O1			
Main storage 4	U	U	U	U	O3	U	U	O1	I1	O1	E1	E1	U	U	O12	O12				
Storage - small parts 5	U	U	U	U	U	U	U	U	O2	O2	I2	E2	U	U	U					
Storage - large parts 6	U	U	U	U	O3	U	U	U	E1	E1	U	U	U	U						
Storage for consumer goods 7	U	U	U	U	U	U	U	U	U	U	U	U	U							
Storage for parts shipment 8	U	U	U	U	U	U	U	U	U	I1	I1	I1								
Picking and packing - light spare parts 9	U	U	U	U	U	U	U	U	U	I4	I4									
Pick & pack - medium spare parts 10	U	U	U	U	U	U	U	U	U	I4										
Pick & pack - heavy spare parts 11	U	U	U	U	O3	U	U	U	U											
Packing - elevator accessories 12	I1	U	U	U	O3	E1	I1	U												
Packing of elevators 13	U	O2	U	U	U	I1	U													
Cable handling and packing 14	O3	U	U	U	U	E1														
Elevator transportation from building 15	E3	E3	U	U	U															
Charging stations - forklifts and AGVs 16	U	U	U	U																
Waste handling 17	U	U	U																	
External storage area - goods reception 18	O2	O2																		
External storage - finished hoist cages 19	A2																			
External storage - finished accessories 20																				

Value	Closeness
A	Absolutely neccesary
E	Especially Important
I	Important
O	Ordinary closeness OK
U	Unimportant
X	Not desirable

Code	Reason
1	Material flow
2	Convinience
3	Movement of equipment
4	Work environment

Figure 15: Relationship Diagram for the Logistics Centre

In the current state, closeness relationships are partly violated due to building separation. In the future state, the relationship matrix is used explicitly to prevent repetition of such spatial misalignments. High-intensity material flows (Reason 1: Material Flow) are prioritised over general convenience.

Particular emphasis was placed on absolute closeness between main storage and picking functions, strong proximity between packing and shipment and minimised distance between goods reception and storage. This directly addresses high transport intensity, fragmented storage and repeated handling.

A relationship flow diagram was also created to visualize the closeness needs, see *Figure 16*, where the numbers correspond to the number for each activity in the relationship diagram.

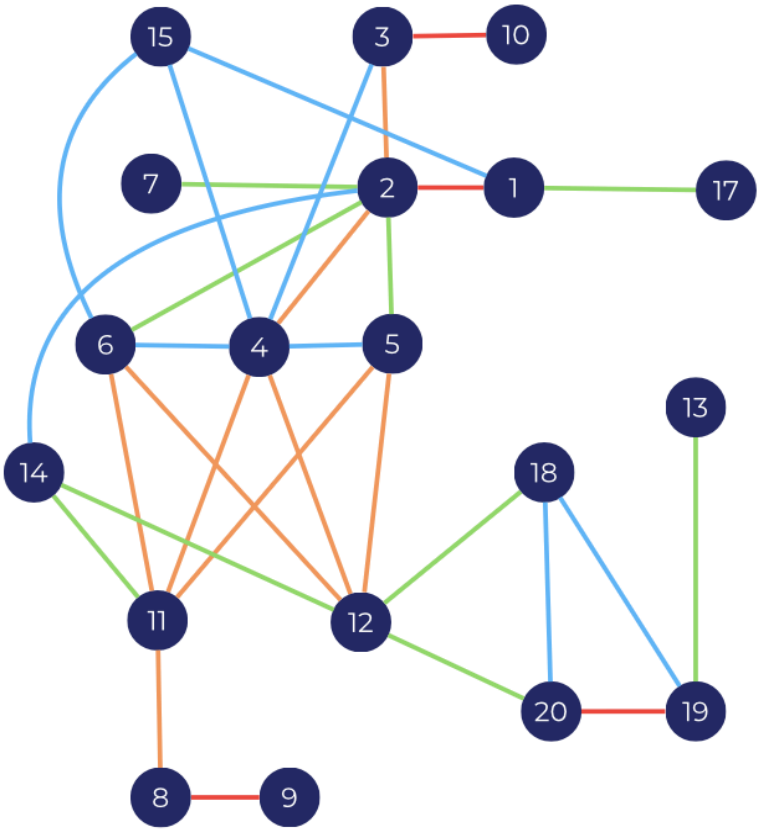


Figure 16: Relationship Flow Diagram for the Logistics Centre

Step 6: Development of Space Relationship Diagram

For step six, a space relationship diagram was developed, which shows the area required for each activity as well as the desired closeness of each activity, see *Figure 17*. The diagram provided a visual representation of how activities could be positioned relative to each other while respecting both spatial constraints and prioritised relationships.

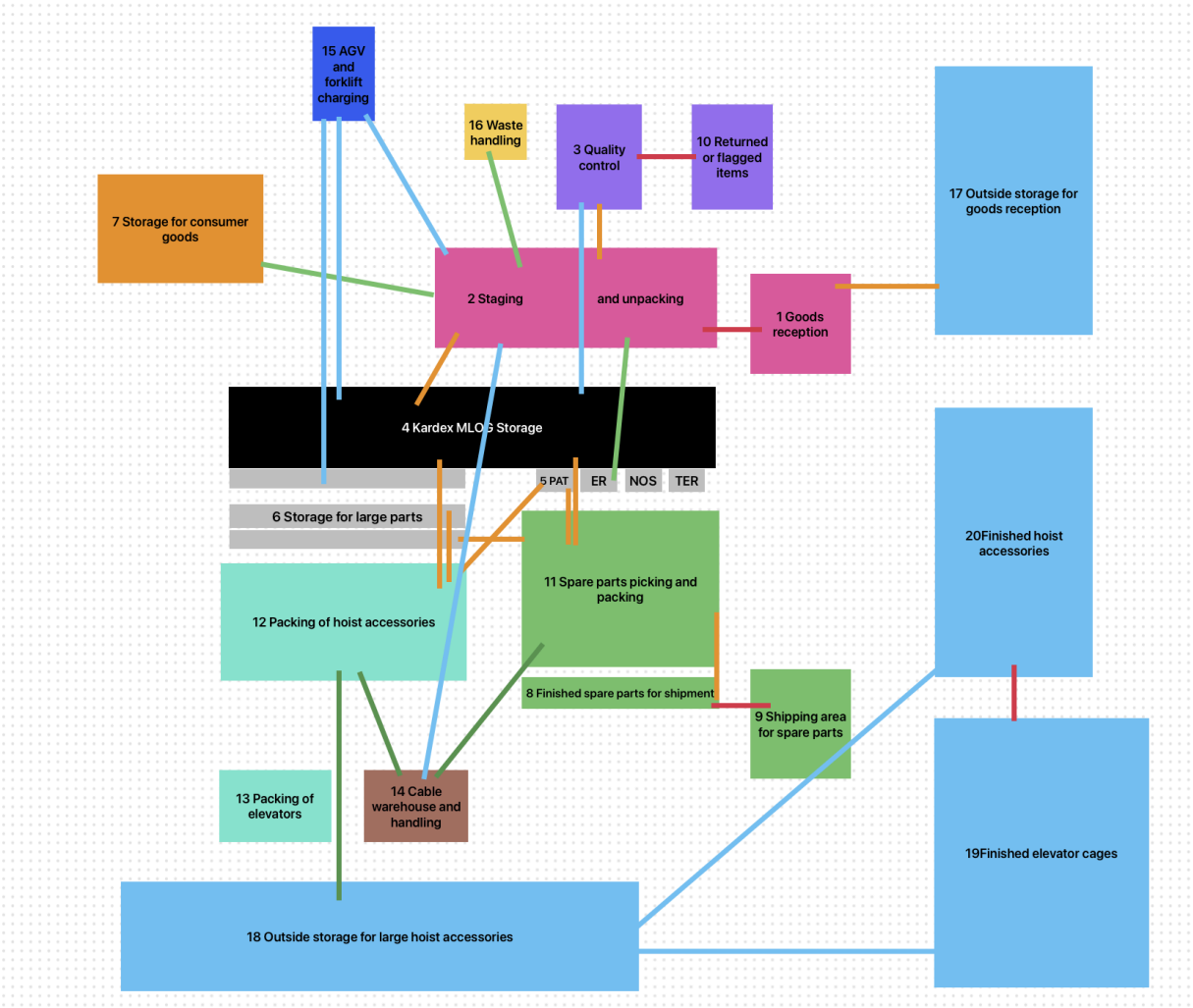


Figure 17: Space Relationship Diagram for the Logistics Centre

Unlike the current configuration, where storage systems are fragmented across buildings, this diagram ensures clustering of storage-related functions. This structural grouping is intended to reduce replenishment loops and improve inventory visibility.

Step 7: Generating Layout Alternatives

To generate layout alternatives, the space relationship diagram was used as a foundation, to build upon. Since the different coloured lines in this diagram correspond to an actual length, meaning that all the green lines are approximately the same length. This enables the layout alternatives to directly reflect the space relationship diagram. The layout alternatives had to be adjusted, which Muther & Hales (2015) clearly defines as a normal part of the process. Multiple layout options were generated through both creative ideation and flow-principle-based design during a brainstorming session. The generation phase deliberately explored different flow logics to evaluate how each configuration influences transport distances, crossing flows, AGV routing feasibility and separation of pedestrian and vehicle traffic. Seven main layout alternatives were picked based on how well they respected the space relationship diagram as well as by discussing with Alimak's Production Engineering Manager. The seven options can be seen in

Figure 18. The different departments are colour-coded in the figure, where the goods reception is shown in pink, the storage for consumer goods in orange, spare parts in green, packing of hoist accessories and elevators in cyan and quality control in purple. The blue areas are the charging stations for AGV and forklift. Yellow is supporting services, brown is cable storage, black is the main storage, white is the storage for large parts and the paternoster storage systems for small parts. Light grey is floor and darker grey show the aisles for forklift and AGV.

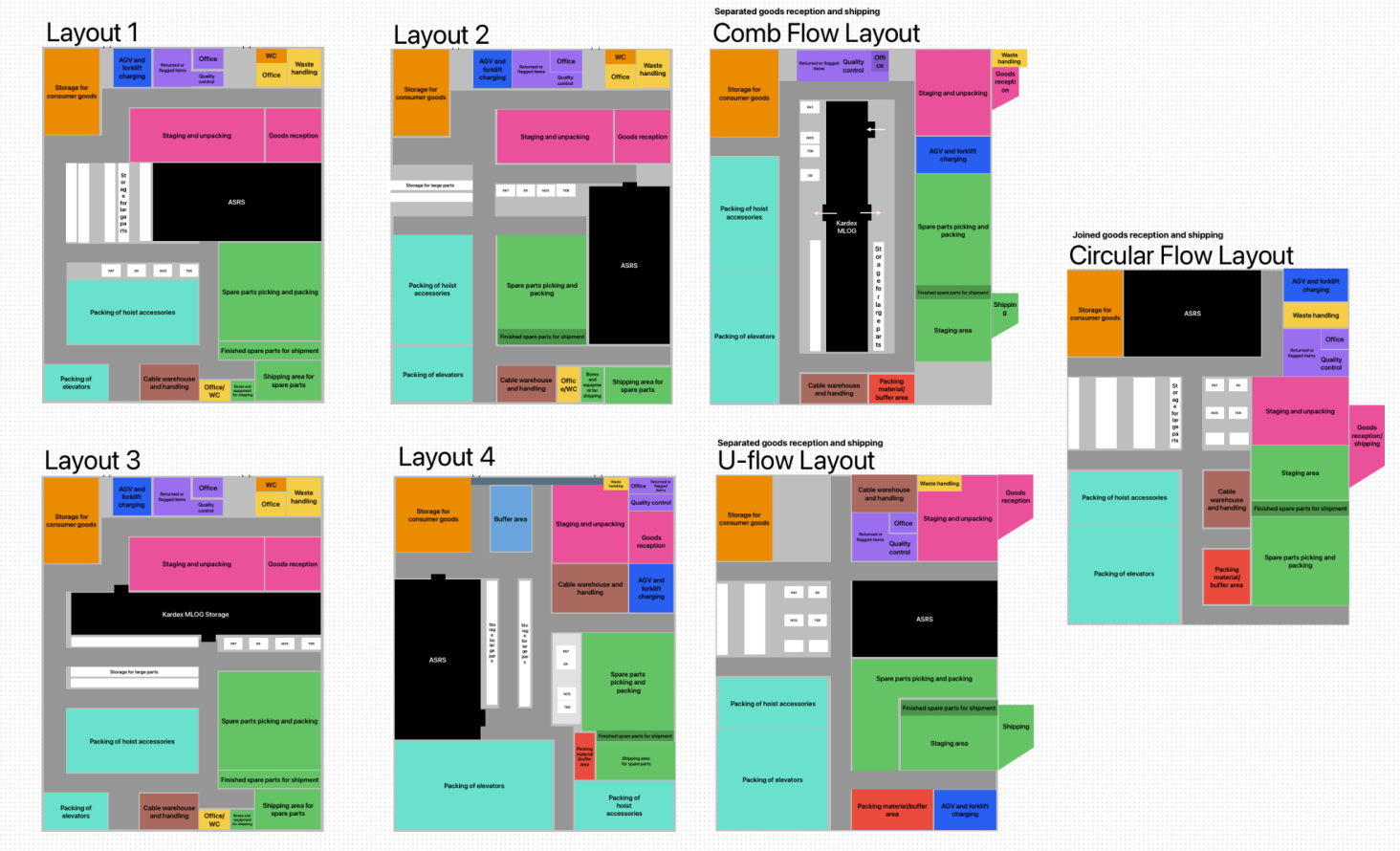


Figure 18: The Layout Alternatives

Step 8: Evaluation and Selection of Layout Concept

With the alternative layout options, *Step 8* could be executed by comparing the options to how well they fulfill the requirements for the logistics centre. The purpose of the evaluation focused on identifying the most suitable flow structure rather than finalised spatial arrangement. The specification of requirements was created and then verified by Production Engineering Manager Mattias Karlsson to validate the requirements for Alimak and the logistics centre. The weight of each requirement has numerical value between three and five, depending on its importance. These have been determined by evaluating what the main goals are for the logistics centre and what is the most important to fulfill. The different options have then been assigned a value to how well they fulfill each requirement with a scale from one to three. The weights were decided as three to five to decrease the effect of different values assigned, since a scale from one to three would indicate that the value assigned has a higher importance than it realistically does. The specification of requirements with weights assigned can be seen in *Figure 19* and a summary of the different areas that the specification of requirement includes can be seen in *Table D*.

Code	Demand	Weight	Layout 1	Layout 2	Layout 3	Layout 4	Comb Flow	U-Flow	Circular Flow
F1	Minimise transport distances between sequential processes	4	2	3	2	3	2	2	2
F2	Support high transport frequency without bottlenecks	4	2	2	1	3	3	1	1
F3	Reduce risk of congestion in transport routes	4	3	2	2	3	3	2	2
F4	Minimize material handling effort and number of handling steps	5	3	3	3	3	3	3	3
C1	Activities with A/E/I relationships are positioned in close proximity	4	3	3	3	3	3	3	3
C2	Undesirable adjacencies (X relationships) are avoided	3	3	3	3	3	3	3	3
C3	The layout supports logical process sequencing	5	2	3	2	3	3	3	3
C4	Connections to external departments are maintained efficiently	4	2	2	3	3	2	3	3
SU1	Layout fits within building geometry and structural constraints	4	3	3	3	3	2	2	2
SU2	Provides adequate space for all activities and supporting services	5	2	2	2	3	3	3	3
SU3	Allows accessibility to workstations, storage and transport areas	4	2	3	2	2	2	3	2
SU4	Enables flexibility for future production rate increases or layout adjustments	3	2	2	2	2	2	2	2
SE1	Separation of pedestrian and vehicle flows where possible	5	2	3	2	2	3	3	3
SE2	Minimised conflict points between transport routes and work areas	4	2	3	2	3	3	3	2
SE3	Provides sufficient visibility in transport corridors and intersections	5	2	3	2	3	3	3	3
SE4	Ensures access for emergency routes and safety zones	5	3	2	2	2	2	2	3
SE5	The layout considers work environment and minimizes physical load	5	3	3	3	3	2	2	3
Total (%)			**80.37**	**88.58**	**76.26**	**92.24**	**86.76**	**84.93**	**85.84**

Figure 19: Filled out Specification of Requirements

Table D: Summarised Specification of Requirements

Category	**Demand fulfilment Layout 1**	**Demand fulfilment Layout 2**	**Demand fulfilment Layout 3**	**Demand fulfilment Layout 4**	**Demand fulfilment Comb**	**Demand fulfilment Circular**	**Demand fulfilment U-flow**
Flow efficiency	84.3	84.3	68.6	100	92.2	68.6	68.6
Closeness relationships	81.3	91.7	89.6	100	91.7	91.7	100
Space utilisation	75.0	83.3	75.0	85.4	77.1	85.4	77.1
Safety and ergonomics	80.6	93.1	73.6	86.1	86.1	86.1	94.4
Total (%)	**80.4**	**88.6**	**76.3**	**92.2**	**86.8**	**84.9**	**85.8**

The evaluation explicitly weighted flow efficiency, closeness relationships, space utilisation, safety and ergonomics. From the specification of requirements, the results are that *Layout 4* performs the best, followed by *Layout 2* and *Layout Comb*. Layout 4 performed highest overall because it minimised crossing flows, clustered storage and picking, created a clear material directionality and allowed structured AGV and forklift routing. This means that the chosen layout is *Layout 4,* and will be moving forward to a detailed layout planning as a part of *Phase III.*

5.1.2 Mapping of Phase III: Detailed Layout

For *Phase III*, a return to the start of the SLP was made for the goods reception. *Step 1-8* was repeated, with some changes made to fit the specific case for each block. Thereafter, the remaining areas that do not require a full systematic planning for the layout was made by working together with workers relevant to each area at Alimak. The objective was to prevent internal inefficiencies within functional blocks recreating the fragmentation observed in the current state. Some steps from the SLP were made for some of the blocks, the measures taken for each activity block will be described below.

Goods reception, staging and unpacking

From the block layout, the three areas of goods reception, staging and unpacking are gathered in one area and therefore will be analysed together in *Step 9.* To determine the detailed layout the SLP was conducted again for this area, where parts of the SLP have been tweaked to fit the issue at hand.

The first step was to determine the PQRST for this block, which can be seen in *Table E.*

Table E: PQRST for the Goods Reception

Factor	**Description**
P - Products	Big parts, pallet-sized goods, half-sized pallet goods, small parts, consumer goods, outsourced parts
Q - Quantity	Delivery quantities are highly varying, no exact number found
R - Routing	Arrival of goods → goods unloading → unpack → control contents → add goods to digital system → sort goods → transport to storage
S - Supporting Services	Forklifts, AGVs, pallet jack, office areas, printer, bathroom, kitchen area, IFS business system
T - Time	One shift with flexitime from 06:00-08:00 until 15:00-17:00

The second step was to create a Process Chart Diagram for the new goods reception, see *Figure 20*. The incoming goods are loaded onto a conveyor belt which is accessible from outside. The goods have a visible QR-code that is scanned when the goods are placed on the conveyor belt. A small door is opened after the QR-code is read and thereafter taken into the goods reception, continuing on the conveyor belt until it either reaches the pickup point for large parts or cables, which do not need to be repacked. Otherwise, the material goes to a station where the packages are opened and inspected and repacked onto pallets where the pallets containing materials for consumer goods and the main storage are transported on conveyor belts to an AGV pickup point. Small parts are instead moved on conveyor belts directly to the paternoster storage. Goods that need further inspection are transported on conveyor belts to an inspection station for quick inspections and can, if needed, be transported to a quality control area. If the goods are checked and good to use, they will be transported to the correct storage system, otherwise it will be transported to a shelving space for flagged or defect material, either to be shipped back or discarded.

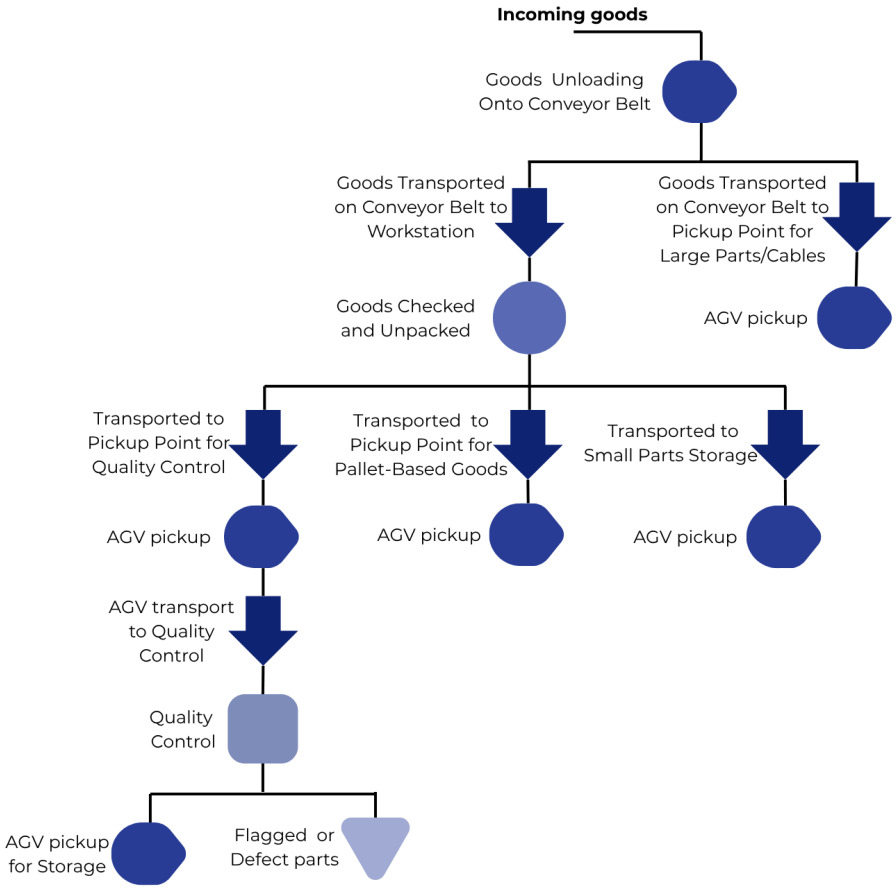


Figure 20: Process Flow Chart for the Goods Reception

The third step was to determine included activities and the required space for goods reception, which can be seen in *Table F.* The space requirements were decided through discussions with workers, through interviews and by analysing the current allowances for each area.

Table F: Defined Activities and Space Requirements for the Goods Reception

Activities	**Space requirements**
Area for truck (external)	8*15 m^2
Goods reception area	6*4 m^2
Staging area	50 m^2
Unpacking and redistribution of goods on pallets	50 m^2
Area for pallets ready for storage	30 m^2
Office areas	20 m^2

No relationship diagram was created, instead closeness was determined by conversations with the workers in the goods reception.

The layout was then determined by brainstorming with the Production Supervisor and some workers from the goods reception. The detailed layout eliminates the need for intermediate staging (goods square) and integrates unloading, inspection, and preparation for storage in a linear flow structure. This reduces unnecessary re-handling and queue formation.

For the activities left in the logistics centre, where the layout was not designed with SLP, the layout and the needed equipment was determined by conversations with workers and by analysing the current layout alongside future needs.. The layouts for each activity were then discussed and verified with workers. The different activities that need to be included for each department can be seen in the tables below.

Table G outlines the key activities within the quality control department. The required spaces include storage for items awaiting quality control or flagged due to deviations and defects as well as a dedicated storage for quality control equipment. Additional space is needed for shelves to hold returned items, as recommended by a worker during an interview, along with an office area equipped with a desk, computer and printer.

Table G: Activities within the Quality Control Area

Quality Control
Storage for material going to quality control or have been flagged
Storage for quality control equipment
Shelves for returned items
Office space with desk, computer, printer

Table H presents the activities within the cable storage area. Since all cables are stored along walls, the primary requirement is sufficient wall area. Beyond this, shelves are needed for finished orders awaiting delivery to other stations, along with a designated working area where cables can be cut to the required length.

Table H: Activities within the Cable Storage

Cable Warehouse
Wall area for cable storage
Shelves for finished cable orders
Working area

The activities within the storage area for consumer goods can be seen in *Table I.* A paternoster system is required to maximise space efficiency for smaller items, while shelves and lockers are needed for larger goods and clothing. Desks and computers are also necessary to support inventory management and related administrative tasks. Drop and pick zones for AGV are also included.

Table I: Activities within Storage for Consumer Goods

Storage for consumer goods
Paternoster storage system for small goods
Shelves and storage for larger goods
Desks and computers
Drop/Pick zones for AGV

Table J covers the activities within elevator packing, also including packing of doors and panels as well as extra material. Shelves for panels are required close to the workstations, with larger batches stored outside and smaller batches kept indoors. Storing smaller batches indoors improves process efficiency, as it eliminates the non-value-adding time associated with defrosting materials during the winter months. Similarly, shelves for frequently used materials support the extra material packing process. A larger floor area is also required for crating the cages. A drop zone for AGV is also included, though is it rare that material from the other storage systems are needed at these stations.

Table J: Activities within Elevator Packing, Doors and Panels Packing and Packing of Extra Material

Packing of Elevators and Doors
Shelves for panels
Floor area for working
Shelves for frequently used materials
Drop zone for AGV

Table K outlines the activities within the packing of hoist accessories A large floor area is needed for the work itself, along with a dedicated work bench at each station to hold tools and other necessities, Drop and pick zones for AGV material delivery are also required.

Table K: Activities within Packing of Hoist Accessories

Packing of Hoist Accessories
Floor area for working
Desk for each station
Drop/pick zones for AGV

Finally, *Table L* details the activities within the packing of spare parts. A sufficient floor area for working is required, along with a workbench for the stations for tools. Drop and pick zones for AGV material delivery are also needed. Additionally, placing a scale at each station for weighing packages would reduce non-value-adding time and further streamline the process. A shared area with computers and printers is also necessary to support administrative tasks.

Table L: Activities within Picking and Packing of Spare Parts

Picking and Packing of Spare Parts
Floor area for working
Desk for each station
Drop/pick zones for AGV
Computers
Scale for weighing packages
Area with computers, printers

Together, these tables formed the foundation for designing the detailed layout, in which each section of the department was carefully considered and planned. The size and proximity requirements for the various activities were then determined through discussions with both production workers and supervisors.

Some details in the layout were decided through the 3D model, since they were highly dependent on unknown variables. These decisions included the amount of forklifts and AGVs and thereby also the amount of charging stations, followed by confirmation on the amount of stations for packing.

The Automated Storage and Retrieval System (ASRS) selected for the main storage area of the logistics centre is a unit load stacker crane ASRS, based on its suitability for high Stock Keeping Units (SKUs), medium throughout environments and its established integration capability with AGV based internal transport systems. Integration with AGVs is achieved through defined drop zones positioned adjacent to the ASRS output stations and at each packing station.

Step 10: Layout Validation Through Discrete Event Simulation

The selected layout was validated using Discrete Event Simulation in Siemens Plant Simulation as described in Section 3.8. The purpose of this step was to determine whether the spatial decisions made during the SLP process translate into viable operational performance, and specifically if the layout can sustain a throughput of 20 units per week under realistic operating conditions. The model was built incrementally, beginning with the most throughput critical subprocess and expanding outward to the full scope of the logistics centre.

Elevator Packing

The model was initiated with elevator packing, as confirming a throughput of eight industrial elevators per week was the primary objective of the validation. Construction hoists do not require packing and flow directly to external storage from final assembly, meaning that only industrial elevators enter this part of the model.

Three simplified elevator types were defined. Small, medium and large, with the occurrence probabilities of 50%, 29% and 21% respectively, controlled via a DataTable. These proportions were determined through discussions with the production supervisor Isabelle Lindgren and are consistent with the current distribution of elevator sizes in production.

Packing time for each elevator type is controlled by a Method giving each elevator type its own triangular distribution of packing times, since the smaller elevators have shorter packing time and the larger kinds have longer packing times. Initial testing with a single packing station showed that the throughout target of eight industrial elevators per week could not be met. Two packing stations were therefore introduced, and the processing time parameters were cut in half to reflect the parallel working stations. The triangular distribution parameters for the packing time of each elevator type can be seen in *Appendix 5*.

Within the same working area, packing of doors and panels as well as extra material run in parallel with the elevator packing. In the simulation, each worker is responsible for a single station, compared to the current state where one worker is responsible for a complete elevator order. Door and panel packing is controlled by a DataTable distinguishing between steel and aluminum panel types, which have different packing time distributions. Doors accompanied by aluminum panels have a likelihood of 80% to occur and doors and steel panels 20%. Extra material follows the same small, medium, large type structure as elevators, with triangular distributions for packing time.

A known simplification in this part of the model is that the elevator type and the corresponding extra material order size are not probabilistically linked. In reality, a large elevator order will generate a large extra material order, but this dependency is not captured in the model. The proportions of each type are correct, but the independence assumption means that the model does not fully replicate the correlation between workloads that would be observed in practice. The triangularly distributed packing times for Doors and Panels can be seen in *Appendix 6*, and for Extra Material in *Appendix 7*.

Hoist Accessories and Spare Parts Packing

To represent the varied material combinations retrieved from different storage systems for each packing order, an assembly table was placed in front of each packing station. This allowed the model to simulate the process by which an AGV retrieves materials from multiple storages before packing order can begin, without requiring a separate model of the full picking process.

A source object connected to each packing station, both at hoist accessories and spare parts, generates pallet entities according to 15 predefined material combinations, each assigned a probability via a DataTable. The 15 combinations were defined to cover the range of order types possible. These combinations can be seen in *Appendix 8.* When a pallet arrives at the assembly table, the model identifies which storage systems are required based on the pallet's combination and triggers the AGV to retrieve

material accordingly. Once all required material units are present on the assembly table, the pallet entity is moved to the packing station where a triangular distribution models variation in packing times.

As with packing of extra material, packing time at hoist accessories and spare parts stations is not linked to order complexity in the model. An order drawing material from four storage systems does not automatically generate a longer packing time than a single storage order. This is a deliberate effort to keep the model easily manageable given the available input data, but it means that the model likely underestimates workload at stations handling the most complex orders.

Entities modeled

The model contains entities representing every product type handled across each storage area, distinguished by colour for visual clarity during model inspection. One entity type represents pallet-base goods retrieved from the main ASRS storage (orange), one represents cables from the cable storage (yellow), one represents small parts retrieved from the paternoster systems (purple) and one represents large parts from the shelving storage (green). The entities are named ASRS_Kit, SmallParts_Kit, LargeParts_Kit and Cables_Kit. Three elevator types are modelled as separate entities, small, medium and large, each with their own processing time distributions and the entity names SmallElevator, MediumElevator and LargeElevator. Two door and panel entity types are modelled, corresponding to doors with aluminum panels and doors with steel panels, named DoorsandAluPanels and DoorsandSteelPanels. Extra material follows the same structure as elevator entities with three size variants, small, medium and large named Extra_Small, Extra_Medium and Extra_Large. The 15 pallet entities used at hoist accessories and spare parts stations each create a unique combination of the storage entity types, allowing order content to vary across runs.

AGV System

Transports through the logistics centre are handled by two worker pools, each representing a distinct AGV type. This pool-based modelling approach was selected because it captures the key behavioural characteristics of AGV systems, including travel time and pick up of materials from different storage types.

The small AGV pool handles retrieval exclusively from the small parts paternoster storage and transports material to the assembly tables at packing stations. The large AGV pool manages all other transport flows between storages and packing stations, including retrieval from the ASRS, large parts shelving and cable storage. Both pools operate at a travel speed of 1.5 m/s with a constant handling time of 2 minutes per pick-up or drop-off interaction. One forklift is modelled for operations outside of the building, responsible for transporting elevator cages and doors from final assembly to the elevator packing station and subsequently to external finished goods storage.

The simulation therefore evaluates throughput, operational availability and bottleneck emergence. This dynamic validation complements the static spatial design and directly addresses Research Questions 3 and 4.

6 Results

This chapter presents the results of the Systematic Layout Planning process and the DES conducted to evaluate the proposed logistics centre concept. The layout evaluation is presented first, followed by the simulation model and its KPI outputs.

6.1 Layout

The proposed logistics centre layout, Layout 4, was evaluated against a weighted specification of requirements covering functional performance, material flow efficiency, work environment, flexibility and safety. The layout achieved an overall requirement fulfilment of 92.2%, indicating strong alignment with the stated design objectives.

Transport distances for the most critical internal flows were reduced by 30%-89% compared to the current state, directly reducing non-value-adding movement and labour requirements. The inbound flow was restructured to eliminate the goods square as well as the intermediate goods square, reducing the number of handling steps per inbound order from between two and five down to a single consolidated receiving process.

The final layout concept for the new logistics centre is presented in *Figure 21*. The layout shows the complete floor plan of the 68m*53.5m building, including placement of all functional areas, workstation arrangements, storage systems and transport corridors for AGV and forklifts.

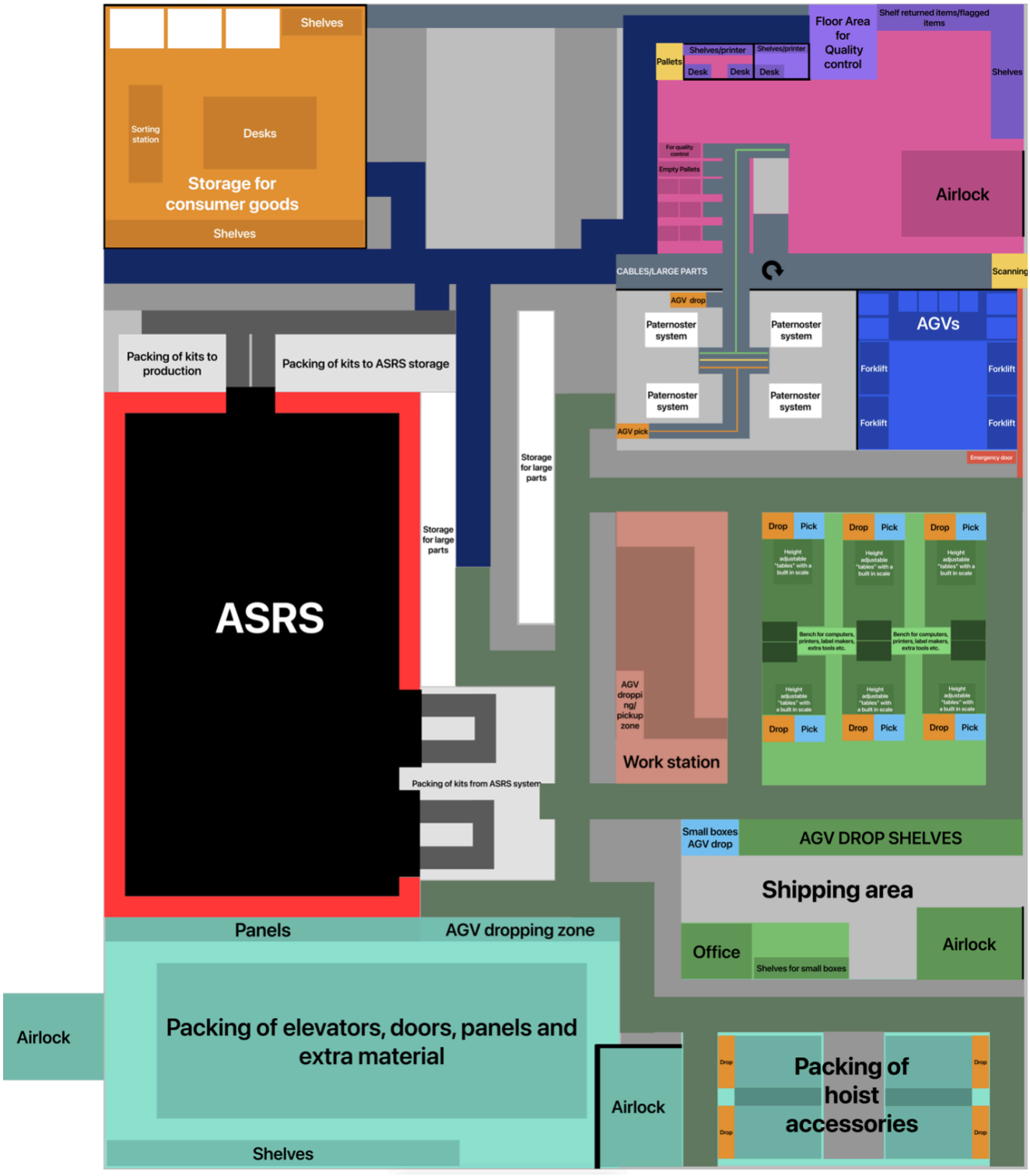


Figure 21: Final Layout Concept for the Logistics Centre

Compared to the current state described in Chapter 4, the proposed layout achieves a reorganisation of spatial relationships. All logistics functions, including goods reception, main storage, spare parts picking and packing, hoist accessory packing, elevator packing and shipment, are consolidated within a single building, eliminating the cross-building transport steps that were identified as the primary source of transport intensity and safety risk in the current state analysis.

The spatial allocation of each functional department within the proposed layout is presented in *Table M*. The dimensions reflect the finalised floor area assigned to each activity in Layout 4 and should be read alongside the final layout concept shown in Figure 21. Where the space requirements analysis in Step 3 and 4 of the SLP (*Table C*) provided estimates based on operational needs and comparisons with the current state, the dimensions presented here represents the resolved spatial allocations after accounting building geometry, AGV, forklift and pedestrian aisle requirements, safety zones and the physical constraints of the 68 x 53.5 m building.

Table M: Dimensions of Departments in Logistics Centre

Department	**Dimensions (m x m)**
Goods Reception	21x15
Consumer Goods	15x14
Small Storage	9x13
Charging Stations	9x9
Cable Storage	6x16
Spare Parts Packing	13x16
Main Storage	16x28
Shipping Area	20x6
Elevator Packing	30x14
Packing of Hoist Accessories	17x8

The largest single allocation is the main storage at 16m x 28m, reflecting its role as the central supply point for all packing stations via AGV-based replenishment. Elevator packing is the second largest are at 30m x 14m , consistent with the requirement identified during detailed layout planning to accommodate elevator packing alongside doors, panels and extra material packing within the same zone. The goods reception occupies 21m x 15m, supporting the restructured conveyor-based receiving flow.

6.1.1 Material Flow in the Proposed Layout

Figure 22 illustrates the inbound material flow in the proposed logistics centre. Before material arrives, the buyers at Alimak orders material on pallets based on which storage system the material is being stored at. On one pallet, only material to one kind of storage system can be delivered. Upon arrival, delivery trucks enter via the dedicated inbound dock on the east side of the building. Goods are unloaded onto a conveyor belt with an access point for both tractors and forklifts. When the goods are placed on the belt, at point 1 in *Figure 22*, a QR code, placed by suppliers, is scanned and specifies the materials and the quantities in the pallet/box. This specifies what storage place the material is to be transported to.

Depending on what type of goods that has arrived it will move in different ways. The most recurring pattern is that material is heading for the main storage, the consumer goods storage or out in the production. This material is transported to the work station at point 2a, where the packages are opened and double checked. At point 3a, different parts of the order will take different routes. The material that is to be stored in the main storage or the consumer goods storage will be transported to either point 4c, 4d or 4e, depending on availability. If the unpacking results in any empty pallets they will be transported on the conveyor belts to point 4b and thereafter transported by an AGV to the pallet storage point at 5a. If any material requires quality control, this delivery will be transported on the conveyor belts to point 4a. Simpler quality controls can be done at this station. If more intricate inspections are needed, for example if the CMM is needed, the material will be transported by an AGV to point 5a. If the material is approved, it will be picked up at point 6b by an AGV and thereafter taken to the correct storage system. If the material is defective, or for any other reason cannot be placed in the storage systems it will be placed on the shelves at point 6a and thereafter handled accordingly. On every pallet, a QR-code is placed by the worker at the workstation that specifies to which storage system it should be transported to, which the AGV then can read.

For small parts that are to be transported to the paternoster storage systems, the same logic applies where the material is transported to the work station at point 2a and checked. However, the boxes in the pallets are separated and placed on the conveyor belt individually, removing the pallet handling needs. The empty pallets are transported to point 4b and handled in the same way as described above. The goods are thereafter transported on conveyor belts all the way to the paternoster systems, to point 4f, removing any transport handling steps. On each box, a QR code is placed by the worker at the work station that specifies the material and to which paternoster system it should be transported to, that will be scanned on the conveyor band on its way to the paternoster systems.

Goods that exceed pallet dimensions, are brought in on the conveyor belt as well, up to 2m wide, and are transported straight to the pickup place for AGV at point 2b. Since the materials that fit these criteria arrive individually on pallets and are visible, there is no need for unpacking at the work station.

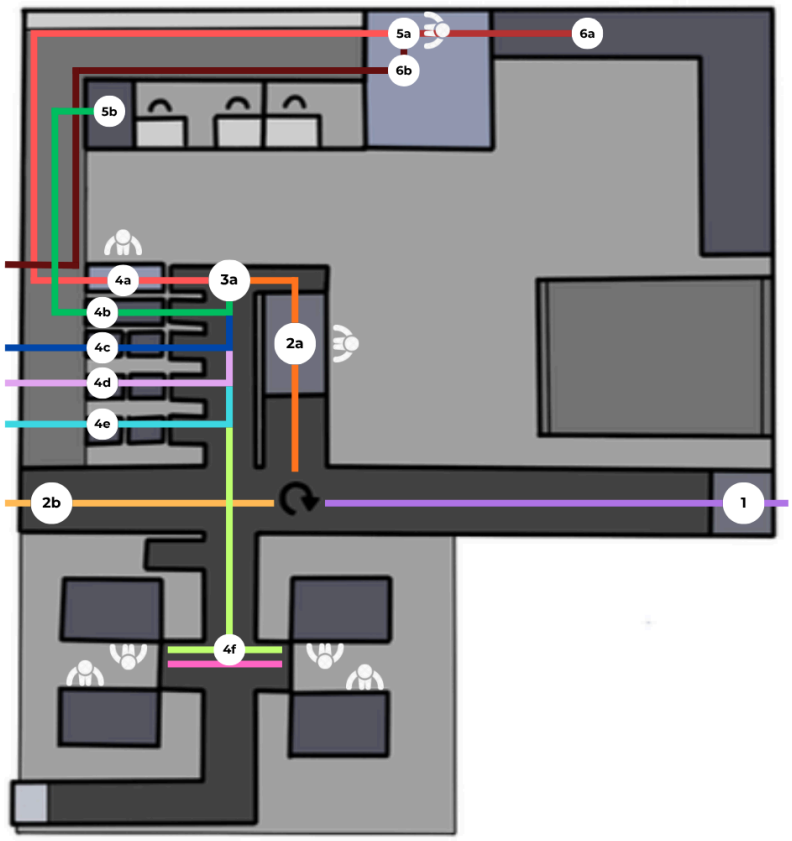


Figure 22: Inbound Flow in the Logistics Centre

Figure 23 depicts the different flows from the AGV pickup zone at the goods reception to the respective storage systems. From point 1, where goods are ready to be transported to the respective storage system, AGVs transport the materials to either point 1a, which is the storage for consumer goods, to point 1b where the goods are placed on another conveyor belt. The goods will move down the conveyor belt until it reaches point 2b, where a workstation is placed. Here, kits are made based on where material is sorted in the ASRS system. Frequently used material is gathered together as well as material that is often recurring in particular order types. Transport to point 1c is large parts exceeding pallet dimensions that are stored on shelves. Point 1d is for AGV transports of cables to the cable storage. In *Figure 23*, point 3 is an exiting flow from the ASRS system, and shows where kits are made that are to be transported to the production, in other buildings on the premises.

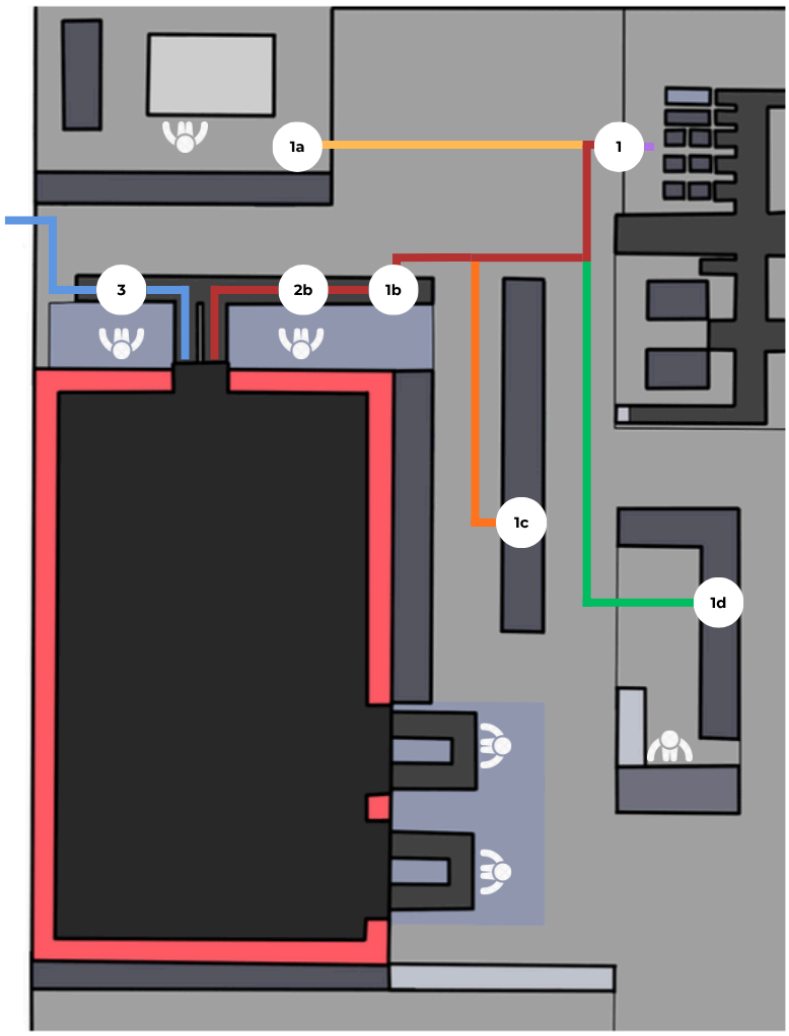


Figure 23: Flow to Storages in the Logistics Centre

Figure 24 shows the distribution of materials from the storage systems in the logistics centre. When a picking order is placed, a signal is sent to the relevant storages to create kits. At the small parts storage, at point 1, the needed material will be packed in a container and thereafter transported to point 1a, where an AGV will pick up the container and transport it to the correct packing stations, either 1b or 1c, which is a simplification in the figure, since it could be any of the spare parts or hoist accessory packing stations. The same principle applies for the rest of the storages. Point 2 shows the large parts storage which will be picked up by a forklift AGV, from the shelves and thereafter transported to the correct station, either 2a or 2b. At the main storage, at point 3, pallets will be sent out where a worker will pick goods to create kits with the needed material from the ASRS storage. Point 3a symbolises the pallets that are sent back into the storage and point 3b is where an AGV will pick up the order and send it to the correct packing station, which can be any of the packing stations, but has been simplified in the figure as point 3c or point 3d. Point 4 shows the cable storage where an AGV will pick up the cables and transport them to the correct station, in the figure, either 4a or 4b.

Point 5 symbolises the door and panel packing station where doors are transported from the final door assembly in the adjacent building and the panels that are stored right next to the packing station. Point 6 shows the packing station of extra material, where the main flow is material from the shelving at the packing station, where outside material is brought inside for easy access. Point 7 shows the elevator packing station, where the elevator is transported from the final assembly.

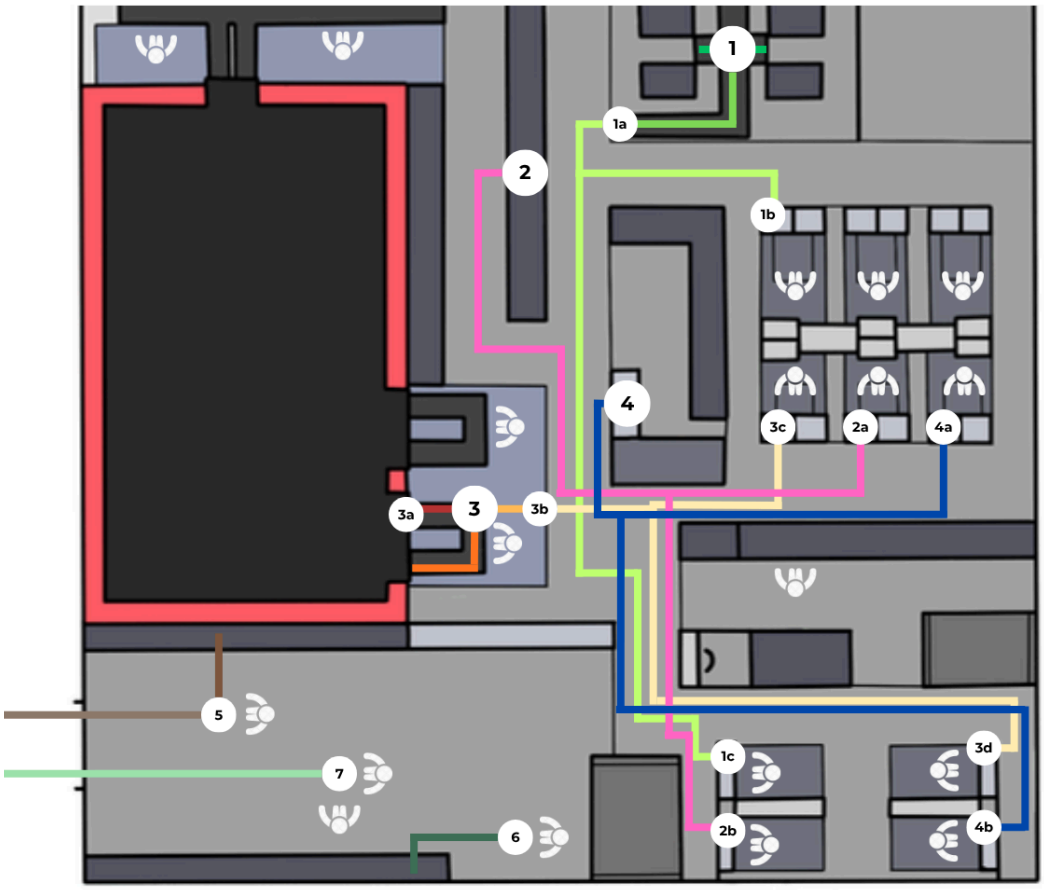


Figure 24: Distribution of Materials from Storages to Stations

Figure 25 depicts the outbound flow for all the packed material in the logistics centre, including spare parts, elevators, doors and panels, extra material as well as hoist accessories. Point 1, 2 and 3 shows the AGV transport to the shelving for finished orders adjacent to the staging area for shipment. The shelves are reachable from both sides which simplifies the AGV routing and minimises crossing traffic. The pallets can thereafter be lifted down in the staging area when it is time for delivery. For smaller orders, that do not require pallet handling, a small goods square, which consists of a simpler shelving area where boxes can be placed for manual handling.

Packed elevator cages, at point 6, doors and panels at point 4 and extra material at point 5 are transported from the elevator packing zone directly to the external finished goods area via the dedicated tractor door, with no requirement to cross internal pedestrian or AGV zones, see *Figure 25*. This separation of heavy

equipment routes from internal logistics directly addresses the safety risks identified in the current state. Hoist accessories are transported directly from the packing area outside from points 7, 8, 9 and 10 to the same external finished goods area as the elevator cages, doors and panels and extra material.

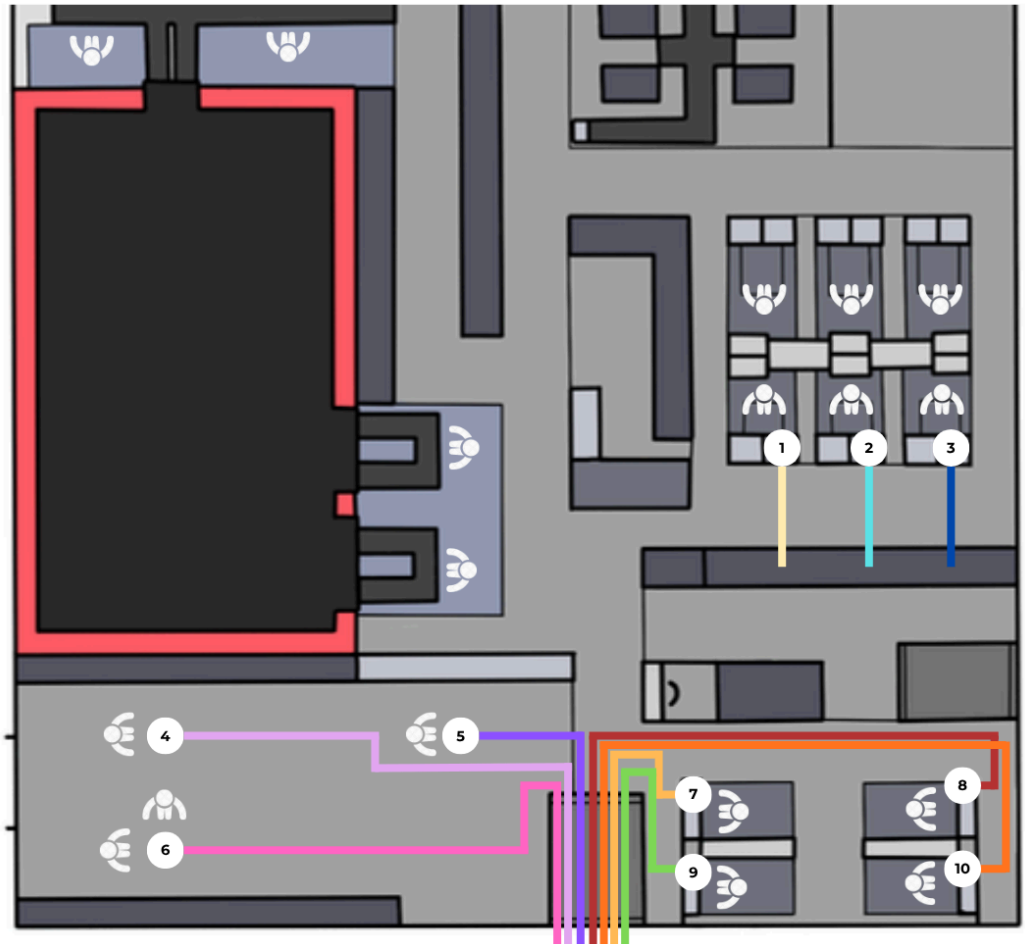


Figure 25: Outbound Flow in the Logistics Centre for Spare Parts, Elevators, Doors, Panels, Extra Material and Hoist Accessories

6.1.2 Comparison of Key Distances

Table N presents a comparison of key transport distances between the current state and the proposed layout. The distances represent typical one-way transport routes for the most frequent material flows.

Table N: Transport Distances in the Current State and in the Proposed Layout

Flow	**Current State (m)**	**Proposed Layout (m)**
Delivery truck route to goods reception	≈350m	≈60m
Goods reception to main storage	≈50m	≈35m
Main storage to spare parts packing	≈225m	≈25m
Main storage to packing of elevators and hoist accessories	≈220m	≈35m

The travelled distance for delivery trucks decreased by 83%, from approximately 350m to around 60m. The distance from the goods reception to the main storage decreased by 30%, from 50m to 35m and from the main storage to the spare parts packing, the distance decreased by 89% and the main storage to packing of elevators and hoist accessories decreased by 84%.

6.1.3 Safety and Ergonomic Improvements in the Layout

The proposed layout directly addresses the safety and ergonomic issues identified in the current state analysis. Delivery trucks are redirected to a dedicated inbound dock at the building perimeter, eliminating the 350-metre internal transport distance and the associated interactions between heavy vehicles and internal transport equipment and pedestrians. Within the logistics centre, AGV and forklift routes are separated from pedestrian zones through designated corridor marking and physical barriers at high risk intersection points, consistent with the EU-OSHA (2001) guidance on vehicle pedestrian separation.

The relocation of elevator packing activities from outdoor to indoor represents a significant improvement in working conditions for workers at this station. The indoor environment eliminates exposure to the extreme temperature variation experienced in northern Sweden, which was identified as a health and safety risk in Section 4.3.4.

The spare parts packing stations are each equipped with a pit-mounted scissor lift table, allowing the working surface to be lowered, parallel to the floor for easy loading and raised to the desired height during packing. This enables each worker to adjust the working height to their individual needs, reducing physical strain during repetitive packing operations. Every station in the logistics centre is equipped with an overhead crane to assist with heavy lifts, minimising the ergonomic risk associated with manual handling of heavy goods.

6.2 Simulation

To evaluate the proposed internal logistics concept, a Discrete Event Simulation model was developed and executed in Siemens Plant Simulation. The model was built and parameterised as described in section 3.8 and Step 10 of Chapter 5. The following sections present the model overview and the resulting KPIs.

6.2.1 Model overview

The simulation model represents the internal material handling system of the proposed logistics centre, covering the AGV-based replenishment of packing station buffers from the central storage areas through to outbound staging. The model operates with one large AGV and one small AGV inside the logistics centre, together with one forklift for outside operations. An overview of the model as rendered in Siemens Plant Simulation is shown in *Figure 26.*

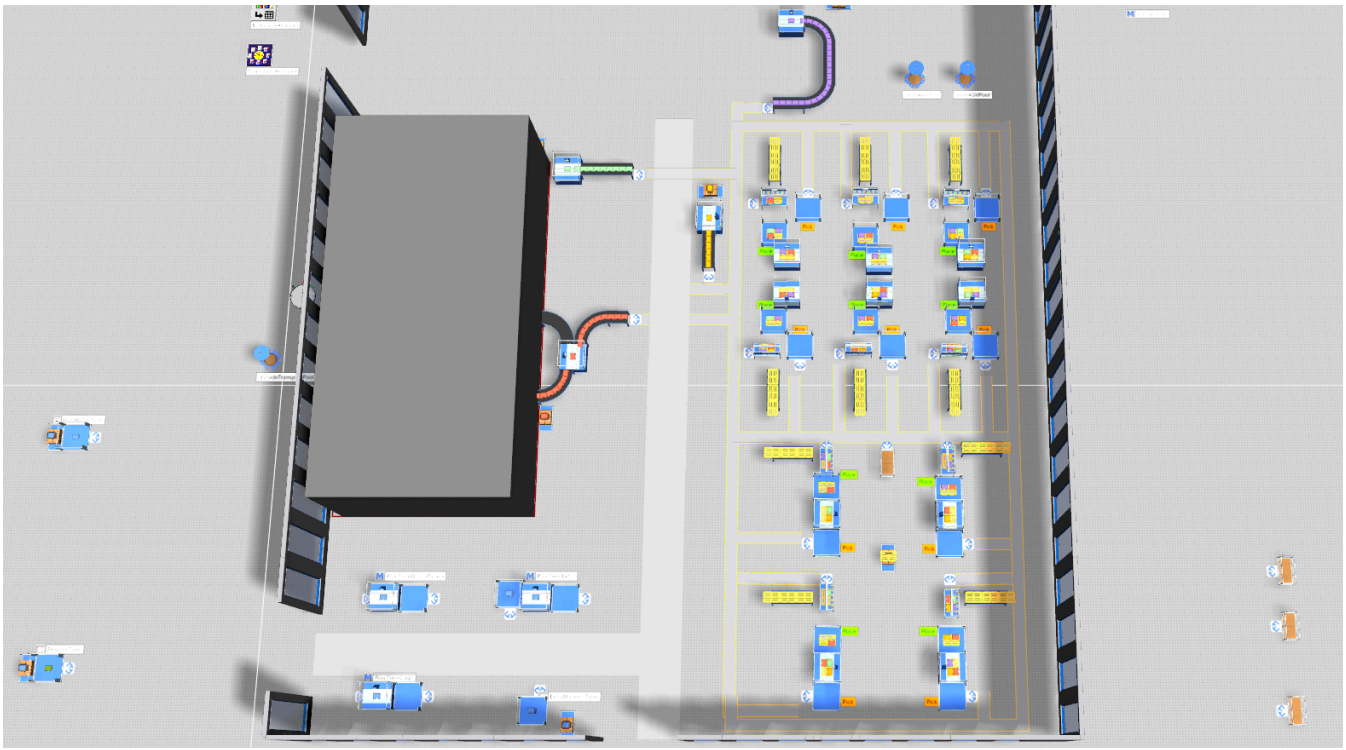


Figure 26: Overview of the Simulation Model in Siemens Plant Simulation

The model structure reflects the spatial layout of Layout 4 directly. AGV path nodes correspond to drop zones as positioned in the detailed layout, and station placements mirror the arrangement established in Phase III of the SLP. This spatial correspondence is what allows the simulation results to serve as a validation of the layout.

6.2.2 KPIs

The KPI results are presented in *Table O*. Operational availability refers to the fraction of total weekly hours during which stations are actively processing, corresponding to a single 8-hour shift, five working days per week, equating to 24% of total weekly hours.

Table O: Simulation KPI Results

KPI	**Target Number**	**Simulated Result**	**Left Interval Bound**	**Right Interval Bound**	**Number of Stations**	**Expected number of stations**	**Operational Availability (%)**
Throughput Industrial Elevators (units/week)	8	11	10.94	11.46	2	1	24%
Throughput Hoist Accessories (crates/week)	15	37	36.67	38.22	(4)	4	24%
Throughput Extra Material (pallets/week)	20	25	25.24	26.45	1	1	24%
Throughput Doors and Panels (crates/week)	8	9	8.21	10.38	1	1	24%
Throughput Spare Parts (boxes/week)	-	41	38.82	45.07	6	6	24%

All product categories meet or exceed their weekly throughput targets. The throughput of industrial elevators reaches a simulated mean of 11 units per week with a target of 8 per week, along with a narrow confidence interval of 10.94 to 11.46, indicating stable and consistent performance across replications. This result was only achievable with two packing stations rather than one, which was established during the model build phase as described in Step 10. Doors and Panels reach a simulated mean of 9 crates per week against a target of 8, with a wider confidence interval of 8.21 to 10.38.

Hoist Accessories show the largest surplus relative to target, reaching 37 crates/week against a target 15. This indicates that two stations might be satisfactory instead of the intended four to reach target levels. Whether this warrants a reduction to two stations or should be retained as built-in future flexibility is discussed in Section 7.2. Extra Material throughput surpasses the target, with 25 units per week,

compared to the target of 20 units per week, with a tight confidence interval of 25.24 units to 26.45 units. Spare Parts has no defined target and is reported for reference only.

Within working hours, no station experiences delays attributable to late material delivery from the AGVs, indicating that the replenishment system keeps pace with demand throughout the operational window. No clear bottleneck is apparent in the results. This should be interpreted with caution, however, as the model generates demand continuously from internal sources without modelling upstream supply constraints such as final assembly readiness or downstream dependencies such as truck scheduling. In a full end-to-end model, bottlenecks would be expected to emerge. This is further discussed in Section 7.2.

The KPIs analysed in Siemens Plant Simulation can be seen in *Appendix 9*.

7 Discussion

This chapter interprets the result presented in Chapter 6, connects the simulation findings back to the layout design and reflects on the strengths and limitations of the methodology applied. The research questions that guided this thesis are addressed in turn. Recommendations for further analysis and implementations are included.

7.1 Does the Layout Meet the Target Requirements?

The proposed logistics centre layout achieved a requirements fulfilment score of 92.2% in the evaluation against the weighted specification of requirements, confirming strong alignment with the stated design objectives. This outcome reflects the effectiveness of SLP as a structured methodology for translating identified inefficiencies into spatial decisions. As Bellgran and Säfsten (2010) argue, layout decisions define how material, information and people interact physically within a production system. The results demonstrate that a systematic approach to those decisions can yield significant improvements over an incrementally developed layout.

The most telling spatial achievement is the consolidation of all logistics functions within a single building. In the current state, functions that are logically sequential in the material flow are physically separated, requiring logistics to compensate for that separation. As Jiang et al. (2023) note, in logistics-intensive environments the cost of movement tends to dominate total operational cost, making spatial alignment with dominant flows a primary design objective. The proposed layout directly addresses this by positioning goods reception, storage, picking, packing and shipment in a coherent sequence, reducing key transport distances by 30-89% compared to the current state.

The restructuring of the inbound flow represents a further significant improvement. The previous multi-step handling process, in which a single inbound order required between two and five handling operations before reaching its point of use, was replaced by a single consolidated receiving flow supported by conveyor belts and AGV-based transports. This aligns with Lean logistic principles emphasising the elimination of non-value-adding transport and handling steps (González-Reséndiz et al., 2018). And with Chen's (2025) findings that optimising receiving area placement and storage system configuration has measurable effects on throughput and operational cost.

Safety and ergonomics were treated as a design parameter throughout the SLP process, consistent with the socio-technical perspective advanced by Bellgran and Säfsten (2010). The relocation of elevator packing activities from outdoor to indoor directly eliminates the health and safety risk associated with extreme temperature exposure in northern Sweden. The separation of delivery truck routes from internal pedestrian and forklift zones addresses the collision and congestion risk identified in the current state analysis, in accordance with EU-OSHA (2001) guidance on vehicle and pedestrian separation. These improvements represent structural changes to the working environment as they provide more reliable risk reduction.

The simulation results further strengthen confidence in the layout. All throughput KPIs met or exceeded their weekly targets and no station experienced material delivery delays during operational hours, confirming that the spatial decisions made during the SLP process translated into viable operational performance under simulated conditions. The combined evidence from the requirements evaluation and the simulation therefore supports the conclusion that the proposed layout concept is well suited to serve as the basis for the detailed design of the logistics centre.

One limitation of the requirements evaluation should be acknowledged. The fulfilment scores were assigned through a combination of structured criteria and engineering judgement and a different assessor working from the same data could produce marginally different scores. This is a known characteristic of weighted decision matrices and is noted by Muther and Hales (2015) as an inherent property of the method. The scores should therefore be interpreted as a structured comparative assessment rather than an absolute measure of layout quality.

7.2 Simulation Performance and Bottleneck Analysis

The DES results confirmed that the proposed layout is capable of sustaining the target production rate of 20 units per week under the modelled operating conditions. This is the primary validation objective of the simulation as defined in Section 3.8, and the outcome provides a dynamic basis for the static conclusions reached through SLP. As Negahban and Smith (2014) emphasise, simulation is particularly valuable for capturing resource interactions and time-dependent behaviour that static analysis cannot reveal. The KPI results demonstrate that the layout's spatial logic translates into effective operational performance when those dynamics are modelled explicitly.

The throughput of industrial elevator packing reached a simulated mean of 11 units per week against a target of eight, with a narrow 95% confidence interval of 10.94 to 11.46, indicating stable and consistent performance across replications. Extra Material throughput exceeded its target by 25% and Hoist Accessories showed the largest surplus relative to target, reaching 37 crates per week against a target of 15. Within operational hours, no station experienced delays attributable to late material delivery from the AGVs, indicating that the replenishment system is well sized for the modelled demand. These results collectively confirm that the layout design creates the conditions for efficient and uninterrupted material flow, which is the central performance objective of the project.

The Hoist Accessories surplus deserves specific attention. At 37 crates per week against a target of 15, the simulation suggests that two stations would be sufficient to meet the target rather than the four stations included in the layout. Whether this warrants a reduction to two stations should be determined by Alimak's production volume forecast beyond the 20-unit target. If the surplus capacity is retained, it provides built-in flexibility for volume growth or temporary demand peaks. If the stations are reduced, the freed space could usefully accommodate a second elevator packing station, which the simulation confirmed was necessary to meet target throughput, or provide buffer area for future operational changes. This decision is further discussed in Section 8.3.

The absence of clearly defined bottlenecks in the simulation results should be interpreted with caution rather than as a confirmation of system robustness. The infinite demand assumption embedded in the

model means that packing stations receive a continuous supply of material without upstream or downstream constraints. In a full end-to-end model, bottlenecks would be expected to emerge, most likely at the elevator packing stations, where packing times are longer, more variable and dependent on final assembly completion. The demand on the main storage would also be higher in practice, as material is supplied to other production buildings on the premises as well as to the logistics centre. The current simulation therefore represents an optimistic scenario for station throughput, and the bottleneck findings should be re-evaluated once a more complete model is available.

7.3 Connecting Simulation Back to the Layout

The use of DES as a validation tool for the SLP-designed layout provided value that static analysis alone could not deliver. The SLP process established optimal spatial relationships and minimised transport distances, but it could not evaluate time-dependent performance, AGV fleet size or station capacity under operational variability. DES captured these dynamic effects, confirming the assumptions embedded in the layout design and providing a concrete quantitative basis for several key implementation decisions.

The most significant design insight generated by the simulation was the requirement for two elevator packing stations instead of the expected singular station. Initial testing with a single station showed that the throughput target could not be met, which directly led to a layout revision. This interaction between simulation and spatial design reflects the complementary relationship between SLP and DES described by Bellgran and Säfsten (2010), and demonstrates how dynamic evaluation can strengthen and refine a statically designed concept. This finding is precisely the type of operational insight that DES is intended to reveal.

The simulation also provided a concrete basis for AGV fleet sizing. Results indicate that one large AGV and one small AGV, together with one external forklift, were sufficient to maintain buffer levels across all stations under the modelled operating conditions. This is likely explained by the relatively long packing times at each workstation, which allow the AGV to complete replenishment cycles before buffers are depleted. In practice, however, a minimum of two vehicles per category is recommended to provide operational resilience. A single vehicle represents a single point of failure (SPOF) where any breakdown, charging requirements or maintenance would immediately halt replenishment for all stations it serves. The simulation confirmed the minimum fleet size for normal operations and the recommendation for two vehicles per category can be seen as a practical implementation margin.

7.4 Limitations of the Simulation

The simulation model confirmed the feasibility of the proposed layout concept under defined operating conditions and provided a reliable basis for the conclusions drawn above. Several modelling assumptions and scope limitations affect the degree to which the result can be generalised to full operational conditions and these should be considered when interpreting the findings.

The most significant limitation is the infinite demand assumption. Packing stations received continuous material input without modelling upstream supply constraints such as final assembly scheduling or downstream constraints such as truck arrival patterns. This means that the simulation represents a best-case scenario for station utilisation and throughput. In a full end-to-end simulation model, irregular

assembly completion, varying delivery schedules and external dependencies would introduce variability that would reduce effective throughput and are likely to expose bottlenecks not visible in the current model scope.

Processing times were modelled using triangular distributions derived from estimates provided by the production supervisor and supported by data from the IFS business system. Triangular distributions are appropriate when sample sizes are insufficient to fit more precise parametric distributions (Law, 2015), and were the most defensible choice given the available data. However, the model's dynamic behaviour is sensitive to the accuracy of the minimum, most likely and maximum estimates. If actual processing times deviate significantly from these estimates, throughput results could differ meaningfully from the simulation's predictions.

Kit-packing time at the ASRS workstation, representing the time an operator spends assembling a kit before AGV collection, was set to a constant five minutes. This is a simplification that may not reflect the variability observed in practice. Kit-packing time directly determines how quickly a buffer is consumed, and therefore how urgently an AGV must respond. Including empirically grounded kit-packing time distributions in a future model would likely increase AGV utilisation and may alter conclusions about fleet size.

The model assumed a single-shift working pattern with no unplanned stoppages or equipment breakdowns. In practice, downtime events would reduce effective throughput and could expose the single-AGV configuration as insufficient. This reinforces the recommendation to deploy at least two vehicles per category in the physical implementation.

Taken together, these limitations indicate that the simulation result represents an operationally optimistic scenario. A conservative interpretation is recommended when using the outputs as a basis for investment or staffing decisions. The model is most useful as a proof of concept for the proposed layout and as a starting point for more detailed modelling as design development progresses.

7.5 Addressing the Research Questions

RQ1: What are the main sources of inefficiency and production losses in the current material flow connected to the new logistics centre at Alimak's production site in Skellefteå?

The main sources of inefficiency identified in the current state analysis were long internal transport distances, fragmented storage across multiple buildings, crossing material flows and outdoor packing operations. Structurally, the current layout reflects incremental development rather than coordinated system design, resulting in a misalignment between process logic and spatial configuration. A single inbound order required between two and five handling operations before reaching its point of use, generating high transport intensity per material unit, elevated labour requirements for non-value-adding activities, increased risk of material damage and exposure to safety risks at crossing traffic points. Outdoor elevator packing exposed workers to extreme temperature variation, representing both an inefficiency and a work environment risk.

RQ2: How can the internal material flow and layout be redesigned to support an efficient logistics centre that supports a production rate of 20 units per week whilst maintaining a good work environment?

The proposed logistics centre layout addresses all identified inefficiencies through a systematic redesign based on the SLP. Delivery truck access was redirected to a dedicated inbound location, reducing the truck route from approximately 350 metres to 60 metres and eliminating interactions with internal traffic and pedestrians. All storage functions were consolidated within a single building using a structured multi-system approach, with the ASRS as the primary storage point for pallet-based goods, paternoster systems for small parts, conventional shelving for large parts and a dedicated storage each for cables and consumer goods. Picking, packing and shipment were positioned in close spatial proximity with structured AGV routing. Elevator packing was relocated indoors, eliminating weather exposure. Pedestrian and vehicle flows were separated throughout the building through designated corridor marking.

RQ3: How does the proposed logistics centre layout perform in terms of throughput, operational availability and bottlenecks when evaluated using Discrete Event Simulation?

All simulated product categories met or exceeded their weekly throughput targets. No station experienced material delivery delays during operational hours and AGV replenishment maintained continuous supply throughout the working shift. No clear bottleneck was identified within the model scope, though this finding should be interpreted with awareness of the model's simplified demand assumption. The results confirm that the layout can support the target production rate of 20 units per week, with the qualification that a more complete end-to-end model is needed before bottleneck behaviour under realistic supply chain conditions can be fully assessed.

RQ4: Which parameters have the greatest impact on the system performance in the simulated logistics centre?

The parameter to which the model was most sensitive was processing time at packing stations, as this drives the majority of the system's dynamic behaviour. Shorter processing times or additional working stations, combined with a proportionally increased AGV fleet, directly increased throughput. The elevator packing station was identified as the most throughput-critical process and the most sensitive to upstream variability, given its longer and more variable processing times relative to other stations. AGV fleet size was a secondary parameter where the simulation indicated that one vehicle per category was sufficient under the modelled conditions, though operational resilience considerations favour a minimum of two.

RQ5: What are the strengths and limitations of combining SLP and DES as a methodology for pre-projection layout design in an industrial logistics context?

The primary strength of combining SLP and DES lies in the complementary nature of the two methods. SLP provided a structured and traceable framework for determining how activities should be positioned relative to each other, grounded in documented flow relationships and operational requirements. DES evaluated whether that spatial logic translated into adequate operational performance under time-dependent conditions. Together, the two methods offered both a static and a dynamic perspective on

the same design problem, which neither could provide alone. The methods are discussed further in Section 7.6.

7.6 Methodological Reflection

In this chapter, reflections are made on SLP as the main design methodology in the project, and on DES as a verification tool. A reflection is then made on the suitability of combining the two methods and lastly, how the methods hold up regarding transferability into other projects.

7.6.1 Reflections on Systematic Layout Planning

The application of SLP as the primary design methodology proved well suited to the pre-projection context of this project. Its most significant contribution was the traceability it provided between identified inefficiencies and spatial decisions. The relationship matrix and space relationship diagram required explicit documentation of proximity priorities, reducing the risk of intuitive placements that could be difficult to justify or revise. As Bellgran and Säfsten (2010) argue, structured layout methodologies are particularly valuable during conceptual production system design because they make the connection between functional requirements and spatial solutions transparent. The expansion of the five conceptual SLP steps into ten operational steps strengthened this traceability further, allowing each activity and its spatial rationale to be treated explicitly and documented in a form that supports stakeholder communication and future revision.

The closeness ratings in the relationship diagram matrix and the space requirement estimates are ultimately judgement-based, which is a known limitation of the SLP acknowledged by Muther and Hales (2015). A different analyst working from the same data could produce a different relationship diagram, which would in turn influence the layout alternatives generated. The weighted decision matrix used for concept selection introduces a second layer of subjectivity in how fulfilment scores are assigned. The selection of Layout 4 is well supported by the structured evaluation, but it should not be presented as a fully objective determination. The quantitative result was complemented by engineering judgement and consultation with Alimak's Production Engineering Manager throughout the process which provides a practical grounding for the decision.

The ten step structure was useful during the block layout phase, where activities and their relationship could be treated systematically. During detailed layout planning, decisions were increasingly driven by physical constraints, worker input and iterative brainstorming rather than formal procedural steps. This reflects a broader characteristic of layout design work, where structured methods provide a valuable foundation in early phases, but detailed design involves a degree of creative and contextual judgement that formal procedures cannot fully cover.

7.6.2 Reflections on Discrete Event Simulation

DES was framed from the outset as a verification tool rather than an optimization tool, consistent with the scope of a pre-projection study and with the recommendations in Bellgran & Säfsten (2010) that simulation should be used to assess whether planned capacities are sufficient before implementation. This framing proved largely appropriate, and the simulation fulfilled its primary purpose of confirming

throughput feasibility and providing a basis for fleet sizing decisions. The model also demonstrated the scalability of the proposed design, showing that the system can be expanded by adding stations or AGVs without structural spatial changes.

The boundary between verification and design was less clear in practice than the initial framing suggested. The most significant example was the elevator packing station capacity. The initial configuration with a single station failed to meet the throughput target, which led directly to the inclusion of a second station. This was not a verification of an existing design decision but a new design decision generated by the simulation. This outcome is consistent with Negahban and Smith's (2014) observation that DES reveals performance effects that static analysis cannot identify, and should be understood as the method working effectively rather than as a deviation from its intended role.

The triangular distributions used for processing times were a reasonable response to limited empirical data and are consistent with Law's (2015) recommendation for bounded asymmetric process time data where sample sizes are insufficient for parametric fitting. The key implication is that simulation outputs should be treated as indicative of the range of likely performance rather than as precise predictions and that collecting empirical processing time data before the facility is built should be a priority for more detailed modelling work.

7.6.3 SLP and DES as Complementary Methods

Taken together, SLP and DES addressed different and largely non-overlapping aspects of the design problem, which supports the claim in RQ5, that the two methods complement each other. SLP structured the spatial logic of the layout, where activities should be placed relative to each other, while DES evaluated whether that spatial logic translated into adequate operational performance under time dependent conditions. Neither method alone would have been sufficient. SLP without simulation would have produced a layout with no basis for assessing AGV fleet size, station capacity, throughput or number of stations. DES without SLP would have required a layout to simulate but would have provided no structured basis for generating or selecting that layout in the first place.

At the same time, the combination has limitations as a methodology. Both methods are resource-intensive, and in a 20-week project the time required to execute the full SLP process and build a functioning 3D simulation left limited opportunity for iteration between the two, which is why the limitation of the simulation model only being for validation was placed in the first place. In an ideal process, simulation results would feed back into layout revisions, which would then be redone to form a cycle that reflects how these methods are used in more extensive industrial design projects. In this project, the integration was largely sequential rather than iterative, meaning that simulation insights such as the needed two stations for elevator packing were incorporated, but a full re-evaluation of the layout against the specification of requirements was not repeated afterwards. Whether Layout 4 would still score highest in such a re-evaluation is likely but therefore not confirmed.

7.6.4 Transferability of the Methodology

The combined SLP and DES methodology applied in this project is broadly transferable to similar pre-projection studies in industrial logistics contexts, provided that certain conditions are met. The methodology is most effective when the system has clearly definable activities with spatial requirements, when material flows can be mapped at the operational level and when sufficient data exists to define a simulation model. It is less well suited to high dynamic or unpredictable systems where flow relationships change frequently or to early stage concepts where spatial constraints have not yet been determined.

For future applications of this methodology, one practical recommendation would be to build the simulation model in parallel with the detailed layout phase rather than sequentially after it. This would allow simulation feedback to inform detailed layout decisions in real time, increasing the degree of integration between the two methods and reducing the risk that simulation findings require layout revisions late in the process.

7.7 Broader Implications and Future Work

The proposed layout and AGV system were designed to support a target throughput of 20 units per week (8 of these being industrial elevators), which is fulfilled. Beyond confirming feasibility at the target rate, the design does accommodate future growth. Adding additional work stations can to a certain extent be done within the building constraints. Introducing a second shift is a further option for increasing throughput without physical expansion. The simulation confirmed this scalability, by reducing process times and increasing the number of AGVs directly increases system throughput.

If production volumes increase significantly beyond 20 units per week, a re-evaluation of the elevator packing station capacity would be needed, given that this station has the longest and most variable processing times and is the most connected to external factors such as final assembly scheduling. For increased demand, focusing on more standardised elevators, and less customer specific orders could also decrease lead time, processing time and storage occupancy, leading to a higher rotation and therefore a higher capacity and production rate.

The methodology developed in this project, specifically the ten-step expansion of SLP integrated with DES-based validation, may also be of value to Alimak for future layout development projects at other sites or for re-evaluation of the logistics centre layout as operational conditions evolve over time. The documentation produced through this project, including the relationship matrix, space relationship diagram and simulation results, provides a structured baseline from which future design iterations can proceed with maintained traceability.

8 Recommendations

Based on the findings of this thesis, the following recommendations are made to Alimak Group Sweden for the implementation of the new logistics centre. The recommendations are structured to address layout implementation, the AGV and ASRS system, workforce transition and areas requiring further investigation prior to implementation.

8.1 Layout Implementation

The proposed Layout 4 concept should be used as the basis for detailed planning. The key spatial priorities established during the SLP, including a dedicated inbound dock at the building perimeter, the ASRS as the central storage point, a linear inbound flow, clear separation of pedestrian and vehicle corridors and indoor elevator packing, should be preserved in any design changes. These features are directly responsible for the efficiency gains identified in this study and should not be compromised for cost reasons without a structured re-evaluation of their impact on material flow performance.

The elevator packing station is a fixed building constraint and should be treated as such. Its exact position must be confirmed before surrounding area allocations and AGV path routing are finalised, as any deviations could have notable effects on the rest of the layout.

8.2 AGV Fleet and ASRS Integration

Two types of AGVs are recommended, one smaller unit for transports from the small parts storage and perhaps also from the main storage. A larger forklift-type AGV for pallet replenishment from the main storage, large parts storage and cables storage is also recommended. Although the simulation indicates that one vehicle of each type is operationally sufficient under baseline conditions, it is recommended to deploy at least two of each type in the physical implementation. The same goes for forklifts for outside use. This provides certainty in the event of breakdown, charging or maintenance requirements and ensures that a single vehicle failure does not halt replenishment across all stations.

The ASRS should be integrated with the AGV path network through clearly defined drop zones adjacent to the storages and packing stations. Storage locations within the ASRS should be logically grouped to reflect picking frequency and order type, with the most frequently accessed items and the most common standard orders assigned to the most accessible storage lanes in order to minimise retrieval cycle times.
AGV and forklift charging stations should be positioned against an outer wall for fire safety reasons, as proposed in the current layout. A more detailed fire safety plan for charging infrastructure should be developed.

8.3 Simulation and Further Analysis

A full end-to-end simulation model that includes upstream constraints such as final assembly scheduling and downstream constraints such as truck arrival patterns should be built before final investment decisions are made. The current model operates under simplified demand assumptions and represents an

optimistic operational scenario. Bottlenecks that did not emerge within the current model scope are likely to appear in a more complete representation of the system.

Empirical processing time data for all packing stations, including kit-packing at storages, should be gathered before building the full model. Processing time is the parameter the current model is most sensitive to, and more accurate input data will significantly improve the reliability and usefulness of simulation results.

The hoist accessories station configuration should be re-evaluated. The simulation indicates that four stations produce considerably more capacity than the 15-crate weekly target. Whether this surplus is seen as flexibility or should be reduced should be decided on Alimak's production volume forecast beyond the 20-unit target. If stations are reduced, the space can be used for extra packing room for elevator packing, needing two stations compared to the expected singular station.

8.4 Workforce and Process Transition

The transition to the new logistics centre represents a significant operational change for the workers at Alimak. The new system introduces AGV-based replenishment, ASRS managed storage, conveyor-based inbound handling and QR code-based goods registration, all of which require new workflows and competencies.

It is recommended that production supervisors from goods reception, spare parts and elevator packing as well as the workers for each department are involved in the detailed planning of workstation layouts, continuing the participatory approach initiated during this project. This involvement ensures that practical operational knowledge is incorporated into the final design and supports staff ownership of the system.

All relevant staff should receive training on the new conveyor and AGV systems and QR code scanning workflows before the logistics centre becomes operational. Training should be completed prior to implementation and should include hands-on practice in the new environment.

8.5 Future Flexibility

The ASRS integration and AGV path network should be designed to accommodate future fleet expansion without structural changes to the building. The simulation confirmed that the system within the current building is scalable without requiring much spatial redesign.

If production volume increases significantly beyond 20 units per week, the elevator packing station should be the first area re-evaluated, as it carries the longest and most variable processing times and is most sensitive to upstream disruption from final assembly scheduling. Increasing the proportion of standardised elevator configurations relative to customer-specific orders would also reduce processing time variability and storage at higher volumes, and is worth considering as a parallel operational strategy alongside physical capacity expansion.

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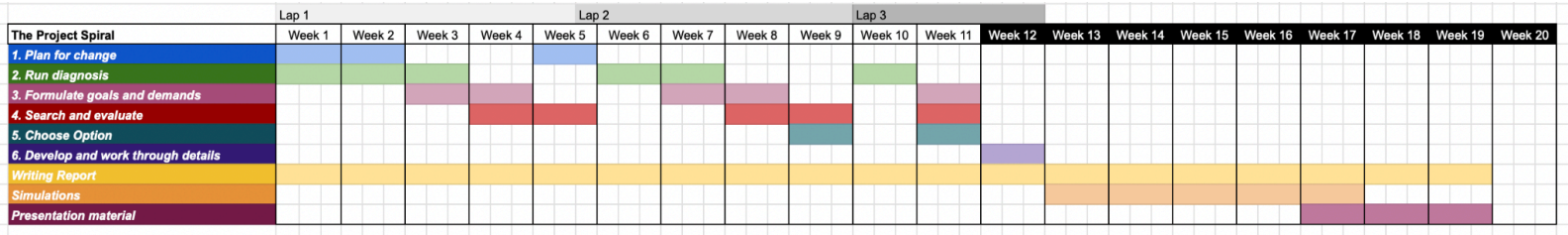
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Appendix

Appendix I: Execution of the Project Spiral, Week by Week



Appendix II: Interview Results with Two Workers at the Goods Reception (Translated by Author from Swedish)

1. **Can you briefly describe your role and your main responsibilities?** Ensuring the right material arrives and is sent to the right place at the right time, including controlling and registering incoming goods with correct documentation to the storage locations.

2. **How long have you worked in the production at Alimak?** 30 years at Alimak, but since October 2025 at the goods reception. Another respondent has worked at the goods reception since 2011.

3. **Can you describe how material moves in this department from when it arrives until it is sent onward?** Small delivery vans deliver manually while larger trucks unload outside and are brought in by tractor. Employees move smaller items inside. Based on experience, workers know what should be stored indoors and outdoors. Items are registered in the IFS business system, assigned a storage location, labelled and placed in the appropriate goods area. Large, medium and small goods are sorted into different areas.

4. **Where do you experience that material often has to wait or is moved unnecessarily?** Some orders arrive without a transport order and must wait for inspection or approval from purchasing. Flagged material is driven in and then back out again. Mast tubes can cause waiting when upstream processes are not completed before delivery.

5. **Are there any flows that you find particularly complicated or inefficient?** The physical paper-based process for workers is inefficient. For each order, the worker must collect a document, walk to the office, return with a newly printed document for storage location and repeat this for every specific order inside every pallet, on every pallet.

6. **Are there any workstations or functions that you think should be located closer to each other?** Quality control should be physically closer to the goods reception, as it is currently far away. It was suggested that one of the three quality control staff could rotate to be stationed at the

goods reception. It would also be beneficial if trucks could reverse directly into the building for unloading.

7. **Are there any functions that absolutely should not be located near each other? Why?** Direct delivery into goods reception without an airlock should be avoided, as it makes the area too cold. Traffic routes for forklifts also need to be considered to avoid conflicts with stored materials.

8. **Where do you experience that material shortages often occur?** Not a significant issue at the goods reception.

9. **Where does too much material or congestion occur instead?** A machine fault can cause goods to pile up in the cold storage area.

10. **Are there any buffers that work well today and should be kept?** No specific buffers were mentioned as particularly well functioning.

11. **Which tools and equipment do you use most in your work?** Pallet trucks, forklifts, tractors, overhead cranes for moving deliveries, computers and printers.

12. **Do you feel that the transport equipment is sufficient today?** Mostly sufficient, though an additional pallet truck could be beneficial to allow working at greater heights.

13. **Are there situations where transport causes disruptions or waiting?** Occasionally a truck may block access when picking up or dropping off goods, but this is rarely more than a 10-minute delay.

14. **Are there places where you experience safety risks related to transport or layout?** Areas around forklifts, particularly when new staff are present. Occasionally difficult interactions with truck drivers can also occur.

15. **Are there any situations where pedestrians and transport vehicles cross each other in an unsuitable way?** Forklifts and tractors sometimes use the same entrance and exit simultaneously. Outdoor forklifts also move around the same area.

16. **If you could change one thing about the layout, what would it be?** A straighter, more direct flow of goods. Ideally, trucks would be able to drive directly into the building and goods would be sorted straight from the truck, or material could be transported on conveyor belts.

17. **What do you think is the most important for the new logistics centre to solve?** A straighter and more efficient pallet flow overall.

18. **Is there anything in today's way of working that absolutely must be kept?** Having the immediate manager nearby is important and should not change. An airlock for deliveries must be maintained to prevent the area from becoming too cold.

19. **Is there anything else you think is important to consider that we have not talked about?** Limited space for large trucks to unload. Clearer instructions for where trucks should stop for safe unloading. Better and clearer outdoor vehicle flow. Implementation of barcode and QR code handling, which would likely improve efficiency significantly. The possibility of digital delivery notes. A shared computer for receiving goods rather than individual ones. The current area has managed a production rate of 22, when more standardised elevators were more common.

Appendix III: Interview Results with One Worker at Spare Parts Picking and Packing (Translated by Author from Swedish)

1. **Can you briefly describe your role and your main responsibilities?** Coordinator for spare parts, responsible for coordinating with other departments such as shipping and sales. Works within a self-managing group and is also responsible for ordering packaging materials.

2. **How long have you worked in the production at Alimak?** 36 years in total, with 15 years specifically in spare parts.

3. **Can you describe how material moves in this department from when it arrives until it is sent onward?** Picking is handled either from the spare parts department's own storage or from the main storage. The aim is to have most items on a Kanban system sourced from other locations. Picking for service vehicles is also managed from this department.

4. **Where do you experience that material often has to wait or is moved unnecessarily?** A lot of unnecessary movement occurs. Some items would be better placed at the department where they are assembled. Electrical components in particular are problematic as having a storage located within the spare parts department would improve efficiency.

5. **Are there any flows that you find particularly complicated or inefficient?** There is a very large number of electrical articles to manage. It would be more efficient if that storage was located closer to or within the spare parts area.

6. **Are there any workstations or functions that you think should be located closer to each other?** The current setup feels reasonably well consolidated.

7. **Are there any functions that absolutely should not be located near each other? Why?** No strong objection, though items that need to defrost should ideally not be stored in the active working area.

8. **Where do you experience that material shortages often occur?** The IFS business system prevents orders from being released if there is a material shortage, so shortages are rarely an issue unless there are inventory balance errors.

9. **Where does too much material or congestion occur instead?** The incoming storage can sometimes become full, depending on prioritisation. When focus is placed on high priority items, other goods may be left waiting, which can cause congestion.

10. **Are there any buffers that work well today and should be kept?** No specific buffers were mentioned.

11. **Which tools and equipment do you use most in your work?** Forklifts, overhead cranes, pneumatic tools, general hand tools and pallet lifters. One counterbalance forklift and one stand-up stacker are available.

12. **Do you feel that the transport equipment is sufficient today?** Generally sufficient, though a scale for heavier and larger items is sometimes missing.

13. **Are there situations where transport causes disruptions or waiting?** No specific issues mentioned.

14. **Are there places where you experience safety risks related to transport or layout?** No currency safety concerns. The situation has improved significantly over time.

15. **Are there any situations where pedestrians and transport vehicles cross each other in an unsuitable way?** The outbound loading area could benefit from more space.

16. **If you could change one thing about the layout, what would it be?** The outbound shipping area is very analogue and paper-based and could be significantly streamlined. Greater digitalisation in these areas would be beneficial.

17. **What do you think is the most important for the new logistics centre to solve?** Improving efficiency and minimising unnecessary forklift driving, as it adds no value to the process.

18. **Is there anything in today's way of working that absolutely must be kept?** The Kanban system works well and should be maintained. While not perfect, it is in a good position. Kanban for electrical components and machine assembly is considered particularly valuable.

19. **Is there anything else you think is important to consider that we have not talked about?** Space and capacity are important considerations. Previously the area was very cramped, which caused frustration and increased safety risks. More space has clearly had a positive impact on the working environment. There is potential to make pallet racking and shelving more space efficient.

Appendix IV: Interview Results with One Worker at Elevator Packing
(Translated by Author from Swedish)

1. **Can you briefly describe your role and your main responsibilities?** Production coordinator for hoist accessory packing, responsible for coordinating and scheduling packing, ordering packing materials, reserving orders, identifying shortages and ensuring elevators arriving are packed on time.

2. **How long have you worked in the production at Alimak?** 37 years in total, with 4-5 years in this specific department.

3. **Can you describe how material moves in this department from when it arrives until it is sent onward?** The main items packed are elevators and drives from assembly. The goal is to have all accessory packing completed by the time the elevator is ready. Parts come from many different storage locations, though most picking is done within their own storage. Items such as masts, doors and anchoring components are stored locally. Nuts and bolts are sent in large quantities from the main storage. Everything is packed within this department.

4. **Where do you experience that material often has to wait or is moved unnecessarily?** Packing of an elevator takes approximately one week before it is ready from the packing station, during which time materials may wait. Items are also sometimes stored in the wrong location, collection from various storage areas across the site.

5. **Are there any flows that you find particularly complicated or inefficient?** No major internal flow issues, though items being stored across multiple external locations requires frequent transport. Moving the main storage loser would reduce transport, but picking from other areas would still be necessary.

6. **Are there any workstations or functions that you think should be located closer to each other?** Storage should be closer to packing, as transport between locations is a significant cost. Snow on outdoor items during winter also causes disruptions, though covering everything with a roof may not be feasible.

7. **Are there any functions that absolutely should not be located near each other? Why?** No concerns raised.

8. **Where do you experience that material shortages often occur?** Shortages are common, typically beginning about one week before an elevator is ready. Electrical components are the most frequent source of shortages, as their delivery timing is not always well coordinated. This can cause packing to halt for several days.

9. **Where does too much material or congestion occur instead?** Congestion is generally not a major issue. Since the packing configuration is fairly similar across most elevator types, the space requirements are relatively predictable.

10. **Are there any buffers that work well today and should be kept?** No specific buffers were mentioned.

11. **Which tools and equipment do you use most in your work?** Primarily forklifts, both indoors and outdoors as well as overhead cranes and carts. Three outdoor forklifts and one indoor forklift are in use. Waiting for forklifts can occasionally occur, as one is frequently operated by the storage operator.

12. **Do you feel that the transport equipment is sufficient today?** Yes, the current equipment feels adequate.

13. **Are there situations where transport causes disruptions or waiting?** No specific issues mentioned.

14. **Are there places where you experience safety risks related to transport or layout?** No safety concerns raised.

15. **Are there any situations where pedestrians and transport vehicles cross each other in an unsuitable way?** No concerns raised

16. **If you could change one thing about the layout, what would it be?** More indoor packing space. Packing outdoors is estimated to take around 20% longer. Currently around 50% of packing is done outdoors, with larger items always packed outside. More indoor space would improve both efficiency and working conditions.

17. **What do you think is the most important for the new logistics centre to solve?** The ability to pack more indoors and in a heated environment, improved internal transport between storage locations and more storage capacity within the building itself.

18. **Is there anything in today's way of working that absolutely must be kept?** Everything currently functions, but there is room for improvement across all areas.

19. **Is there anything else you think is important to consider that we have not talked about?** No additional comments.

Appendix V: Processing Times Elevator Types in Simulation Model

```
param mu : object -> time;


if mu = .UserObjects.SmallElevator then
    return z_triangle(45:00, 30:00, 1:30:00);

elseif mu = .UserObjects.MediumElevator then
    return z_triangle(1:15:00, 45:00, 2:30:00);
else
    return z_triangle(4:00:00, 2:00:00, 6:00:00);

end;
```

Appendix VI: Processing Times Doors and Panels in Simulation Model

```
param mu : object -> time;

if mu = .UserObjects.DoorsAndAluPanels then
    return z_triangle(3:00:00, 1:05:00, 7:00:00);

else
    return z_triangle(4:00:00, 1:05:00, 9:00:00);

end;
```

Appendix VII: Processing Time Extra Material in Simulation Model

```
param mu : object -> time;

if mu = .UserObjects.ExtraForSmall then
    return z_triangle(0:45:00, 0:30:00, 2:00:00);

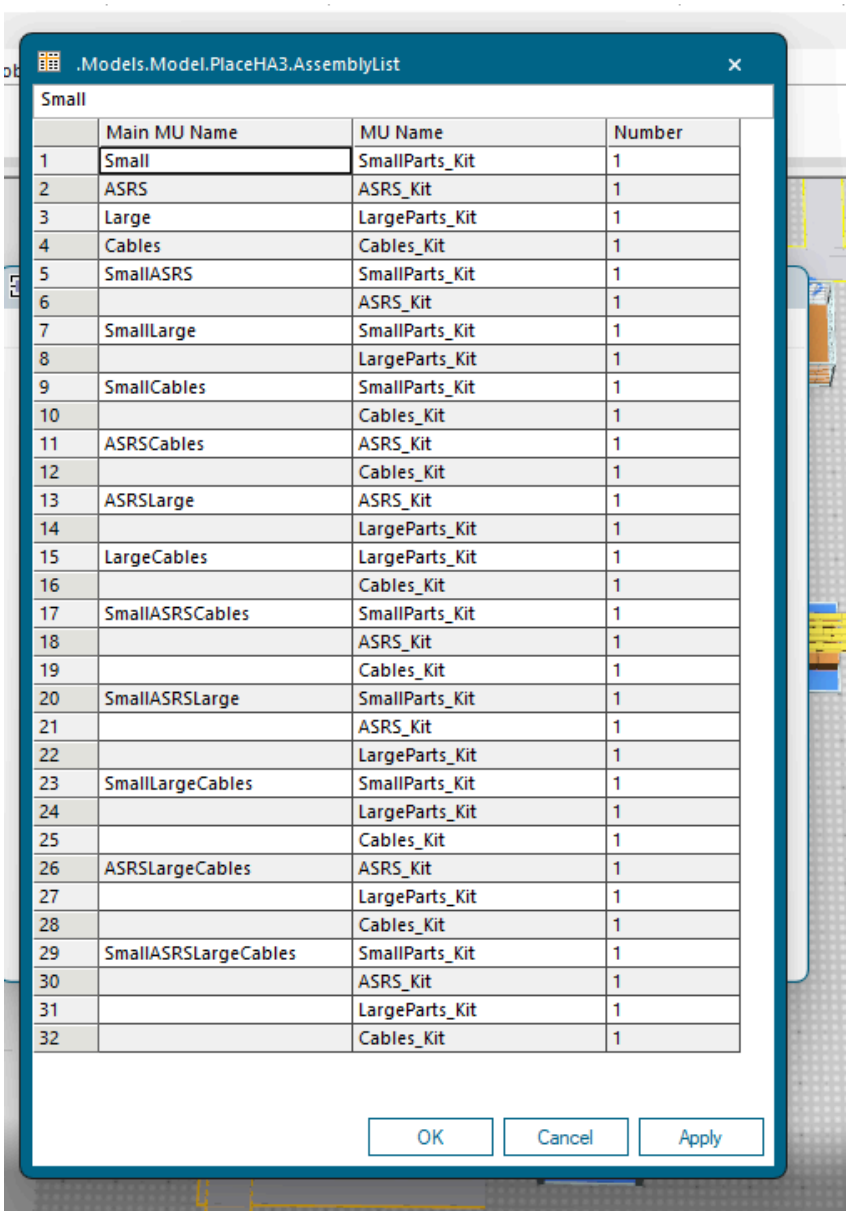
elseif mu = .UserObjects.ExtraForMedium then
    return z_triangle(1:00:00, 0:45:00, 2:30:00);
else
    return z_triangle(1:15:00, 1:00:00, 2:45:00);

end;
```

Appendix VIII: Pallet Entities: Assembly List for Packing Stations in Simulation Model

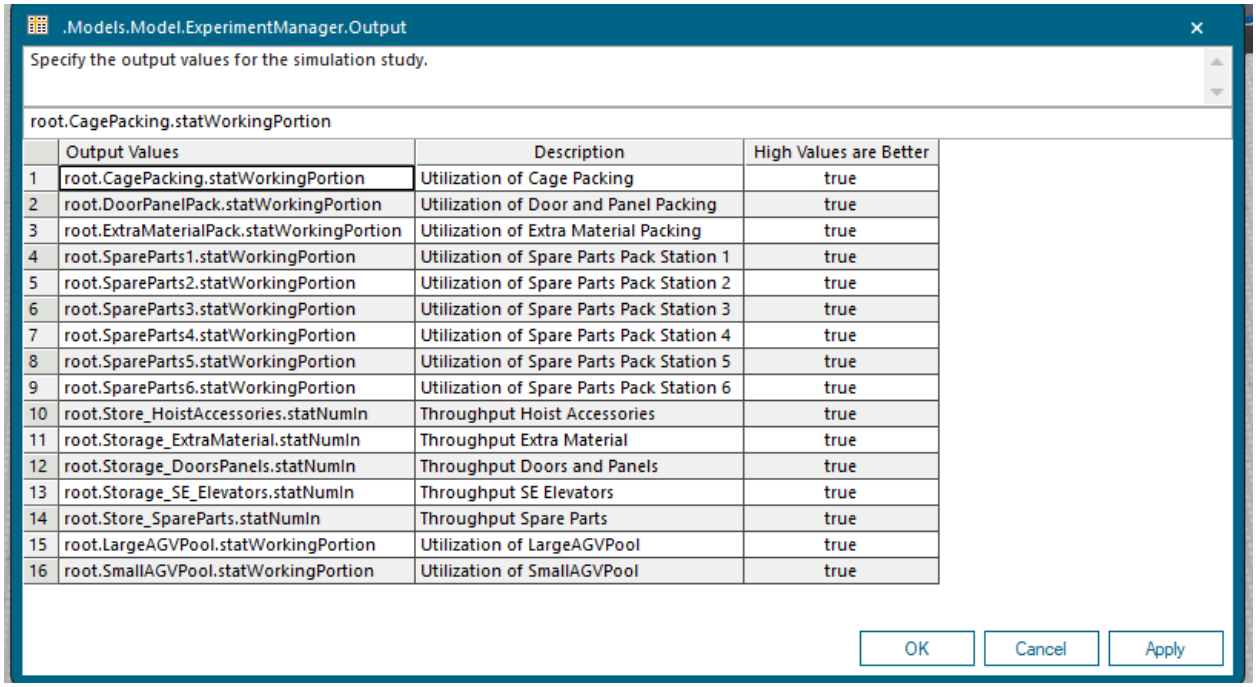
Main MU Name: Pallet based entities

MU Name: Entities from the different storage systems



	Main MU Name	MU Name	Number
1	Small	SmallParts_Kit	1
2	ASRS	ASRS_Kit	1
3	Large	LargeParts_Kit	1
4	Cables	Cables_Kit	1
5	SmallASRS	SmallParts_Kit	1
6		ASRS_Kit	1
7	SmallLarge	SmallParts_Kit	1
8		LargeParts_Kit	1
9	SmallCables	SmallParts_Kit	1
10		Cables_Kit	1
11	ASRSCables	ASRS_Kit	1
12		Cables_Kit	1
13	ASRSLarge	ASRS_Kit	1
14		LargeParts_Kit	1
15	LargeCables	LargeParts_Kit	1
16		Cables_Kit	1
17	SmallASRSCables	SmallParts_Kit	1
18		ASRS_Kit	1
19		Cables_Kit	1
20	SmallASRSLarge	SmallParts_Kit	1
21		ASRS_Kit	1
22		LargeParts_Kit	1
23	SmallLargeCables	SmallParts_Kit	1
24		LargeParts_Kit	1
25		Cables_Kit	1
26	ASRSLargeCables	ASRS_Kit	1
27		LargeParts_Kit	1
28		Cables_Kit	1
29	SmallASRSLargeCables	SmallParts_Kit	1
30		ASRS_Kit	1
31		LargeParts_Kit	1
32		Cables_Kit	1

Appendix IX: Statistics Analysed through Experiment Manager in Simulation Model



	Output Values	Description	High Values are Better
1	root.CagePacking.statWorkingPortion	Utilization of Cage Packing	true
2	root.DoorPanelPack.statWorkingPortion	Utilization of Door and Panel Packing	true
3	root.ExtraMaterialPack.statWorkingPortion	Utilization of Extra Material Packing	true
4	root.SpareParts1.statWorkingPortion	Utilization of Spare Parts Pack Station 1	true
5	root.SpareParts2.statWorkingPortion	Utilization of Spare Parts Pack Station 2	true
6	root.SpareParts3.statWorkingPortion	Utilization of Spare Parts Pack Station 3	true
7	root.SpareParts4.statWorkingPortion	Utilization of Spare Parts Pack Station 4	true
8	root.SpareParts5.statWorkingPortion	Utilization of Spare Parts Pack Station 5	true
9	root.SpareParts6.statWorkingPortion	Utilization of Spare Parts Pack Station 6	true
10	root.Store_HoistAccessories.statNumIn	Throughput Hoist Accessories	true
11	root.Storage_ExtraMaterial.statNumIn	Throughput Extra Material	true
12	root.Storage_DoorsPanels.statNumIn	Throughput Doors and Panels	true
13	root.Storage_SE_Elevators.statNumIn	Throughput SE Elevators	true
14	root.Store_SpareParts.statNumIn	Throughput Spare Parts	true
15	root.LargeAGVPool.statWorkingPortion	Utilization of LargeAGVPool	true
16	root.SmallAGVPool.statWorkingPortion	Utilization of SmallAGVPool	true